Visual disorders

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UNLV MED 808

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Proptosis

- Distended extraocular orbital muscles
- Usually due to Grave's disease
  - Aka Thyroid Ophthalmopathy
  - Muscle fibrosis and excess extracellular matrix is deposited

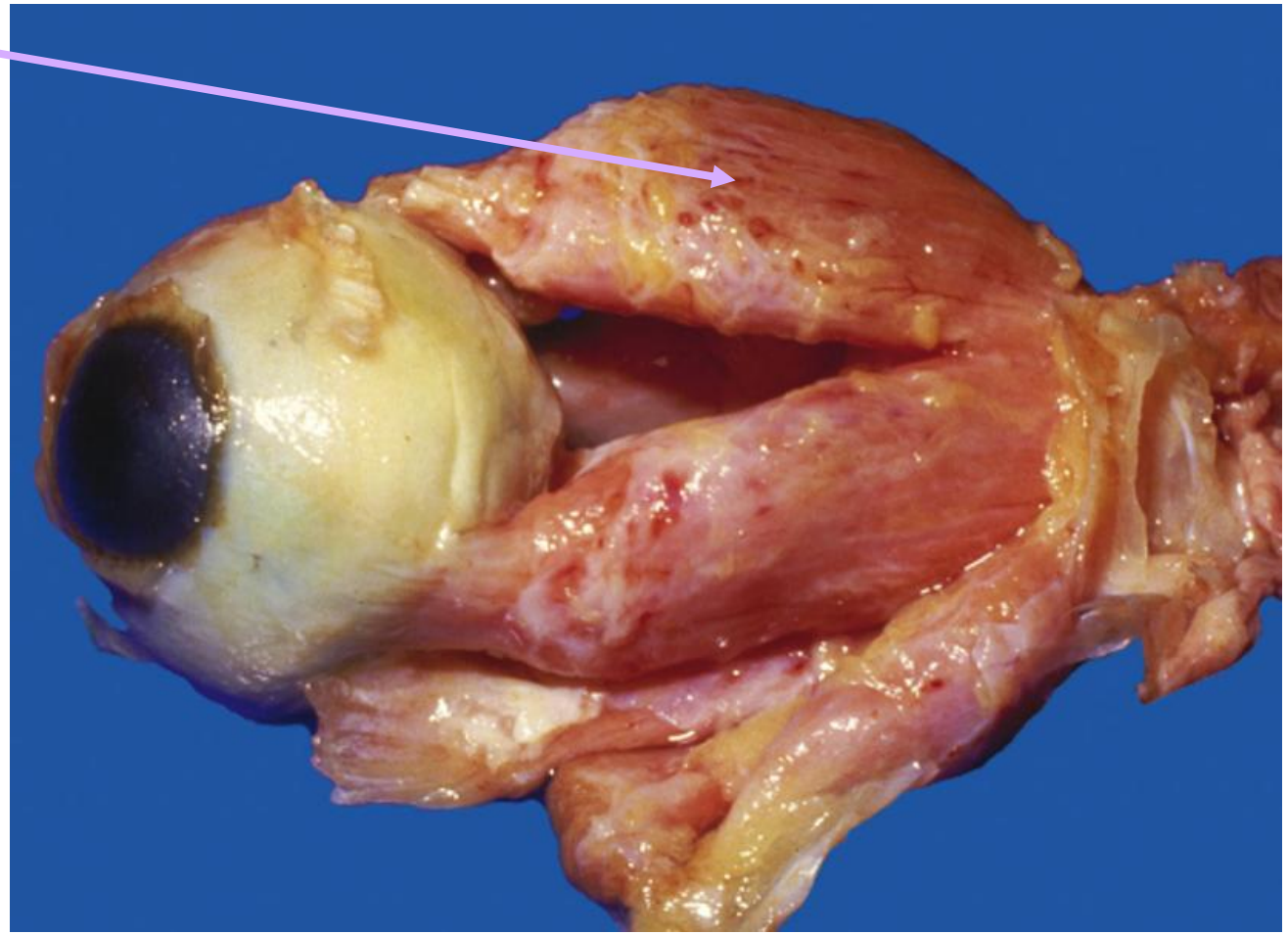


Figure 29.2

Proptosis

- Increase in orbital contents push eyeball forward out of socket
- May lead to corneal drying due to malformation
- Other causes
  - Glioma and meningioma
  - Other causes produce non-classic proptosis
    - Sarcoidosis
    - Pleomorphic adenoma
    - Lymphoma
    - Adenoid cystic carcinoma

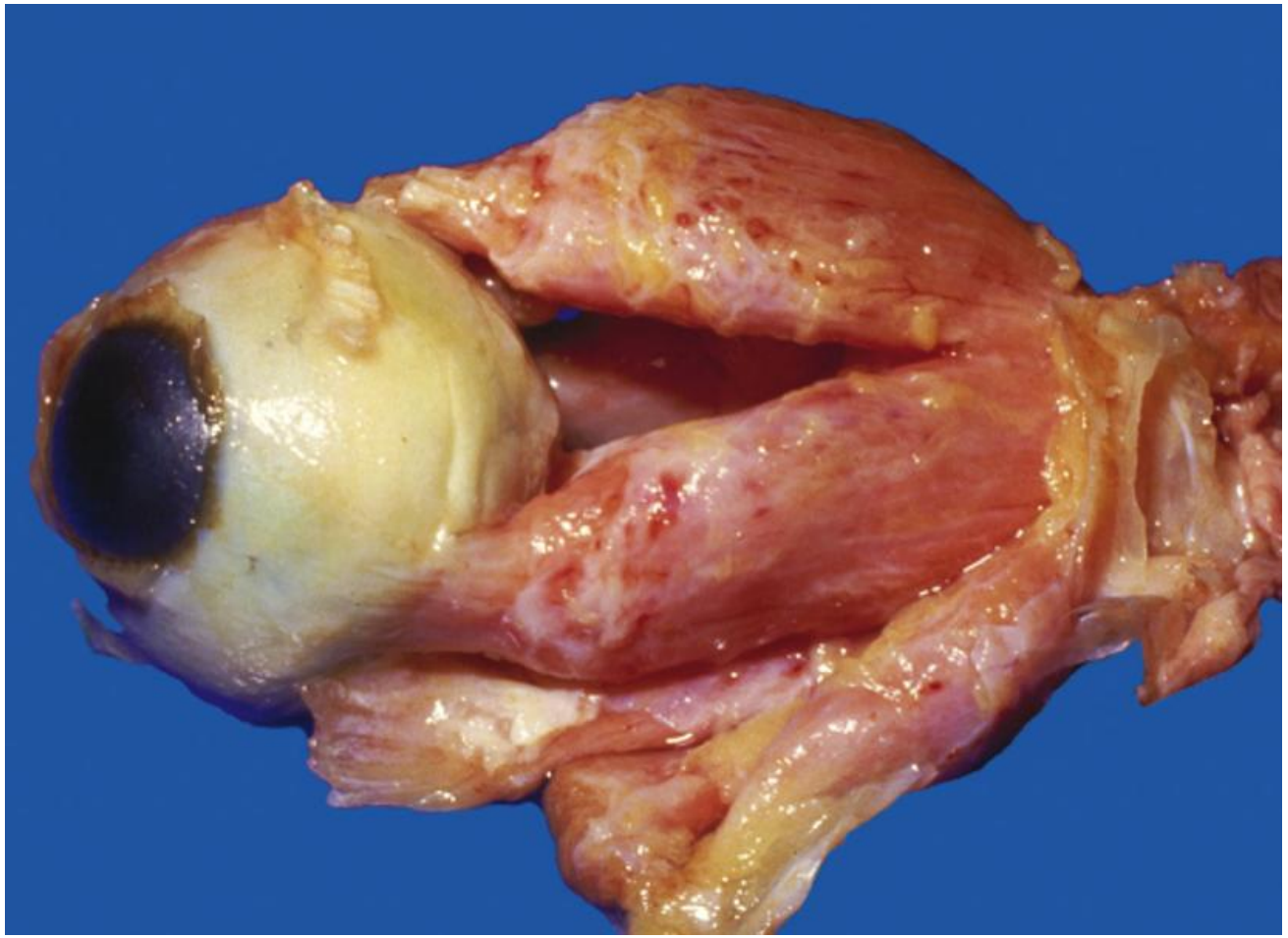
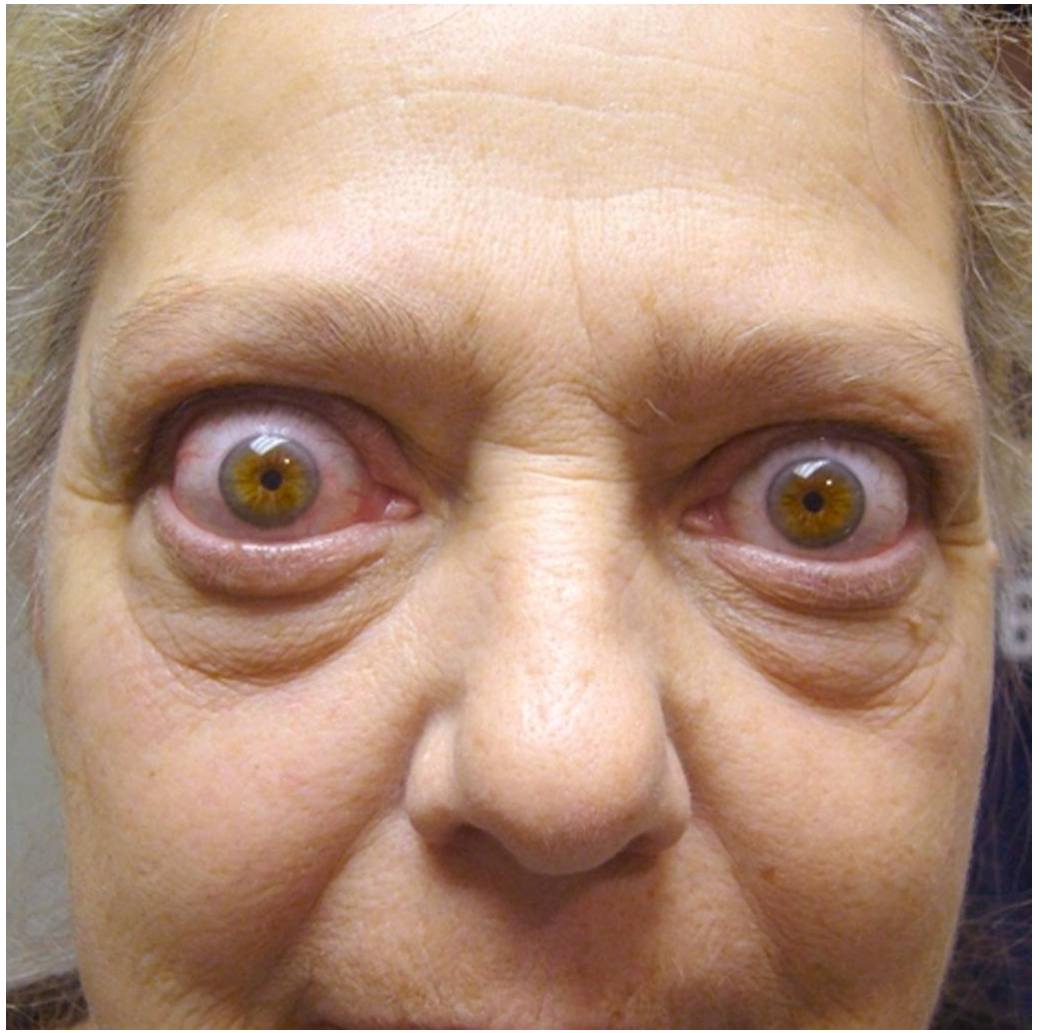


Figure 29.2

Grave's disease

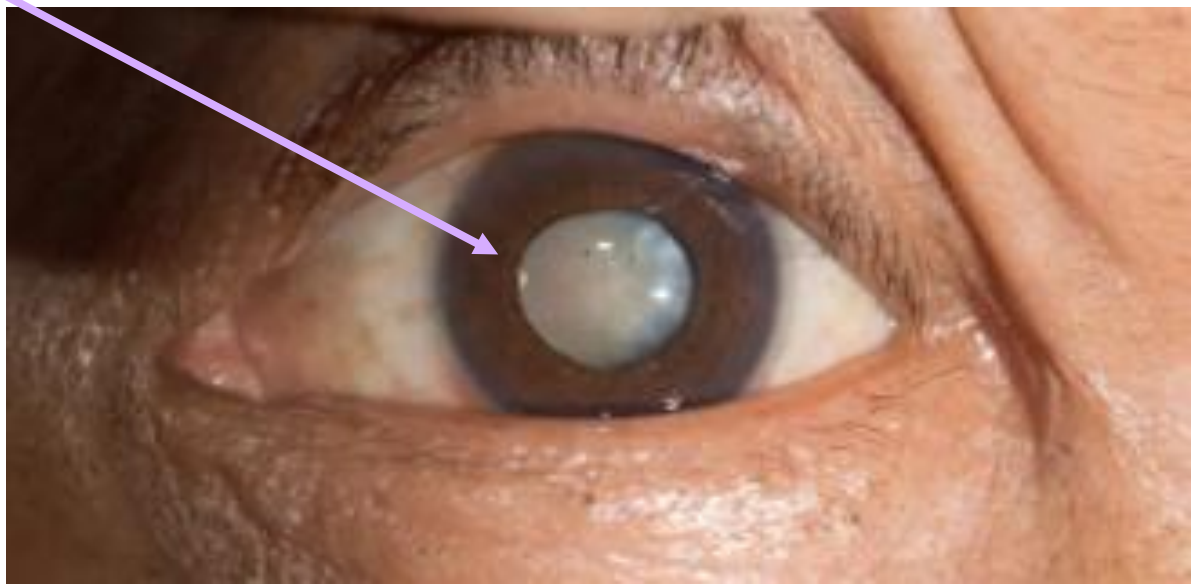
- Autoimmune thyroiditis
- Most common cause of hyperthyroidism
- Prevalence
  - young adults, usually women
- Graves ophthalmopathy
  - Causes of the proptosis:
  - Orbital extracellular matrix
  - Fibrosis of the muscles
    - Cytokines activate TSH receptors on ocular fibroblasts and fat cells



Robbins and Kumar Basic Pathology, Figure 18.8

Cataracts

- Associated with
  - **Diabetes**
  - Other risk factors
    - Galactosemia
    - Wilson disease
    - Atopic dermatitis
    - Corticosteroid use
    - Radiation
    - Trauma
- Treatment
  - Prosthetic intraocular lens after removal of inner lens contents



https://www.aao.org/eye-health/diseases/what-are-cataracts

Cataracts

- Presentation
  - Clinical Symptoms
    - Gradual blurring of vision
    - Glare and halos around lights
    - Difficulty reading, faded colors, and poor night vision
  - Typical Findings
    - Clouding of the lens noticeable during an eye examination

Cataracts

- Early Stage
  - Protein aggregates within lens
  - Mild Clouding, often unnoticed by the patient
  - Slight difficulty with night vision or glare sensitivity
- Intermediate Stage
  - Increased Opacification and accumulation of aggregates with oxidative damage
  - Blurred vision and difficulty reading small print
  - Increased sensitivity to bright light and difficulty with night driving
- Advanced Stage:
  - Severe opacification
  - Worsening protein clumping, calcium deposits, oxidative damage
  - Substantial vision loss with colors may appear faded or yellowed
  - Significant impairment in low-light conditions and reading

Cataracts

- Types
  - Age-related cataract
    - Most common Type
    - Presents initially with myopia (nearsightedness)
    - Nuclear sclerosis
      - Opacification of lens nucleus
      - Urochrome yellowish-brown pigment accumulation distorts blue perception
  - Rarer types of cataracts
    - Posterior Subcapsular Cataracts
      - Migration of lens epithelium at the posterior pole of lens
      - Caused by long term corticosteroid use, diabetes, radiation exposure

Glaucoma

- Group of diseases causing optic neuropathy
- Clinical Symptoms
  - Gradual loss of peripheral vision (tunnel vision)
  - Ocular tonometry and ophthalmologic examination
- Acute cases (angle-closure)
  - Sudden severe pain, headache blurred vision, increased tearing
- Prognosis
  - Progressive condition
  - Controlled by normalizing IOP
    - medications, laser therapy, or surgery
    - vision loss can be halted but not reversed

Glaucoma

- Most glaucoma caused by increased intraocular pressure (IOP)
  - Measured with Ocular Tonometry
    - 19th century tonometer
  - Rare forms caused by blood supply problems to optic nerve
- Two major types
  - Open-Angle Glaucoma
  - Angle-Closure Glaucoma



Albert DM, Keeler R. The Pressure: Before and after Schiøtz. Ophthalmol Glaucoma. 2020 Nov-Dec;3(6):409-413. doi: 10.1016/j.ogla.2020.04.015. Epub 2020 May 5.

Glaucoma

- Vitreous humor creates IOP
  - Produced in ciliary body in posterior chamber
  - Travels through pupil
  - Absorbed in trabecular network in anterior chamber
  - Drained through Schlemm's canal

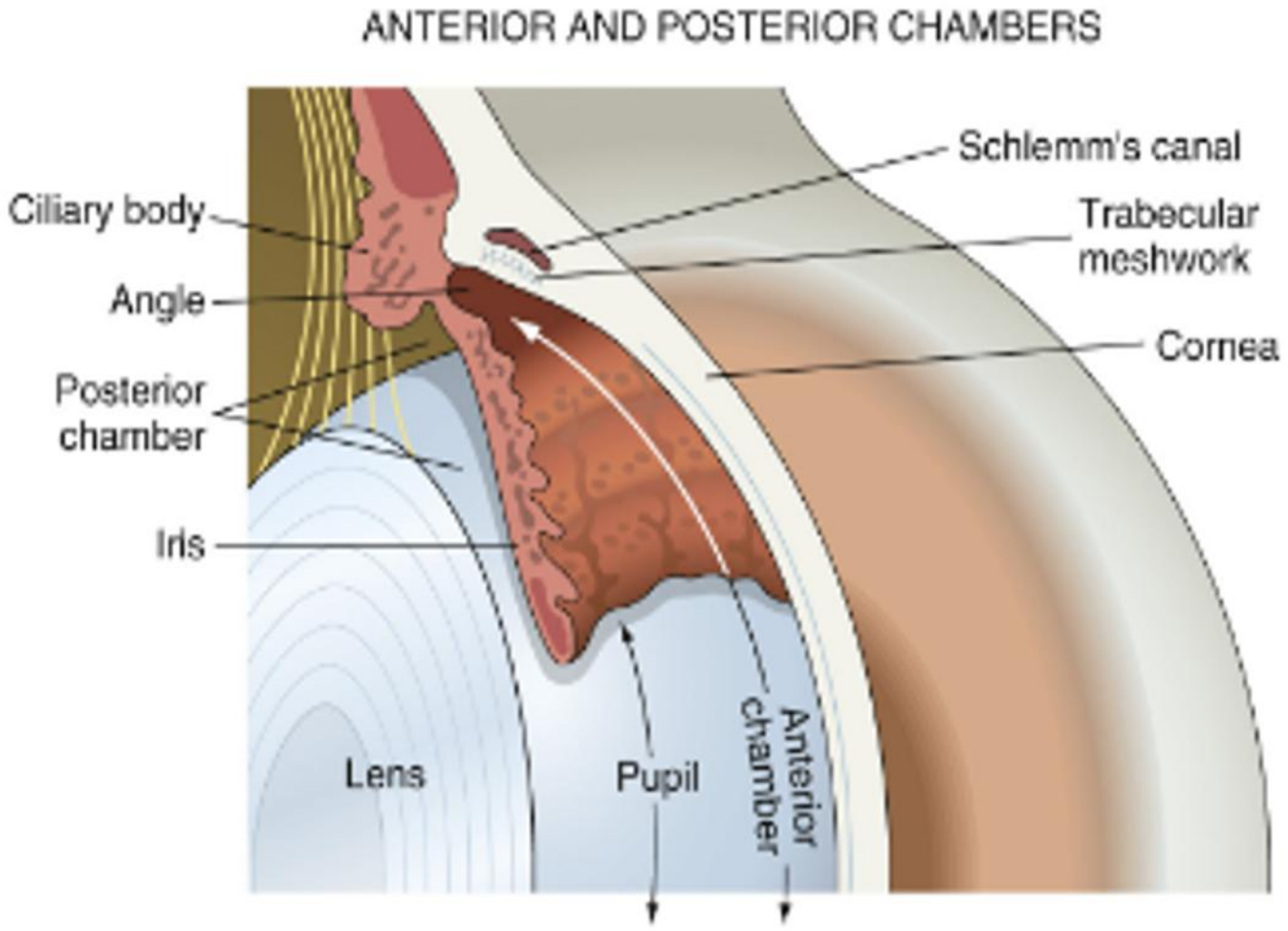


Figure 29.11

Glaucoma

- Open-Angle Glaucoma
  - Usually slow and progressive worsening
    - Often discovered on routine eye exam
  - Primary open-angle glaucoma
    - The most common type
    - Decreased vitreous humor drainage through the trabecular network or Schlemm canal
      - Rare cases may be caused by increased ciliary body production of vitreous humor
  - Secondary forms causes by debris from trauma or tumors

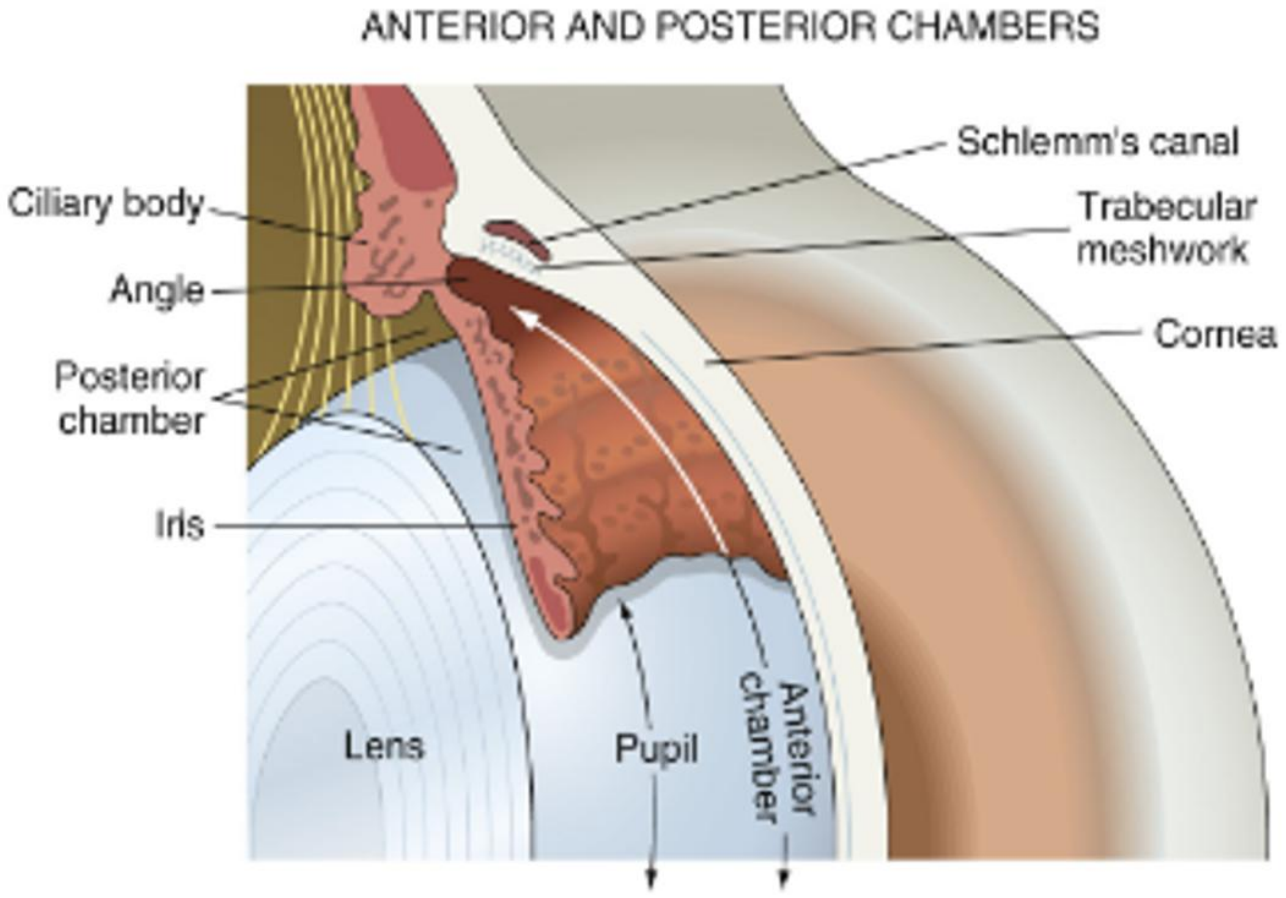


Figure 29.11

Glaucoma

- Primary Angle-Closure Glaucoma
  - May require immediate treatment
    - High risk of rapid vision loss
  - Anterior chamber angle is decreased or closed
    - Blocks drainage of aqueous humor
  - **Hyperopia** (far-sighted patients have increased risk)
    - Iris bombe – iris bows forward
    - This blocks the trabecular network and Schlemm canal

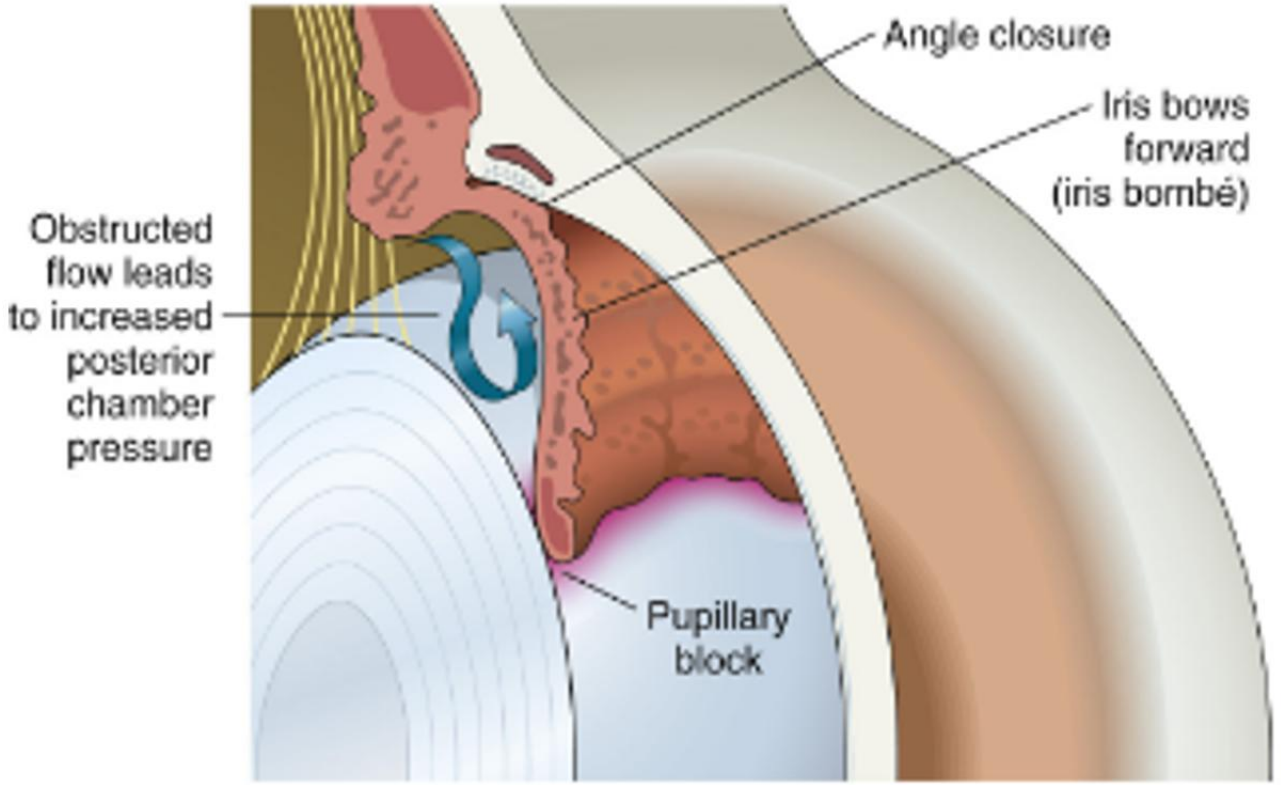


Figure 29.11

Glaucoma

- Secondary Angle-Closure Glaucoma
  - May require immediate treatment
    - High risk of rapid vision loss
  - Anterior chamber angle is decreased or closed
    - Blocks drainage of aqueous humor
  - Usually due to **neovascular glaucoma**
    - Ischemic changes from systemic diseases
    - Membranes forming on iris due to increased VEGF
    - Seen in **diabetes mellitus**

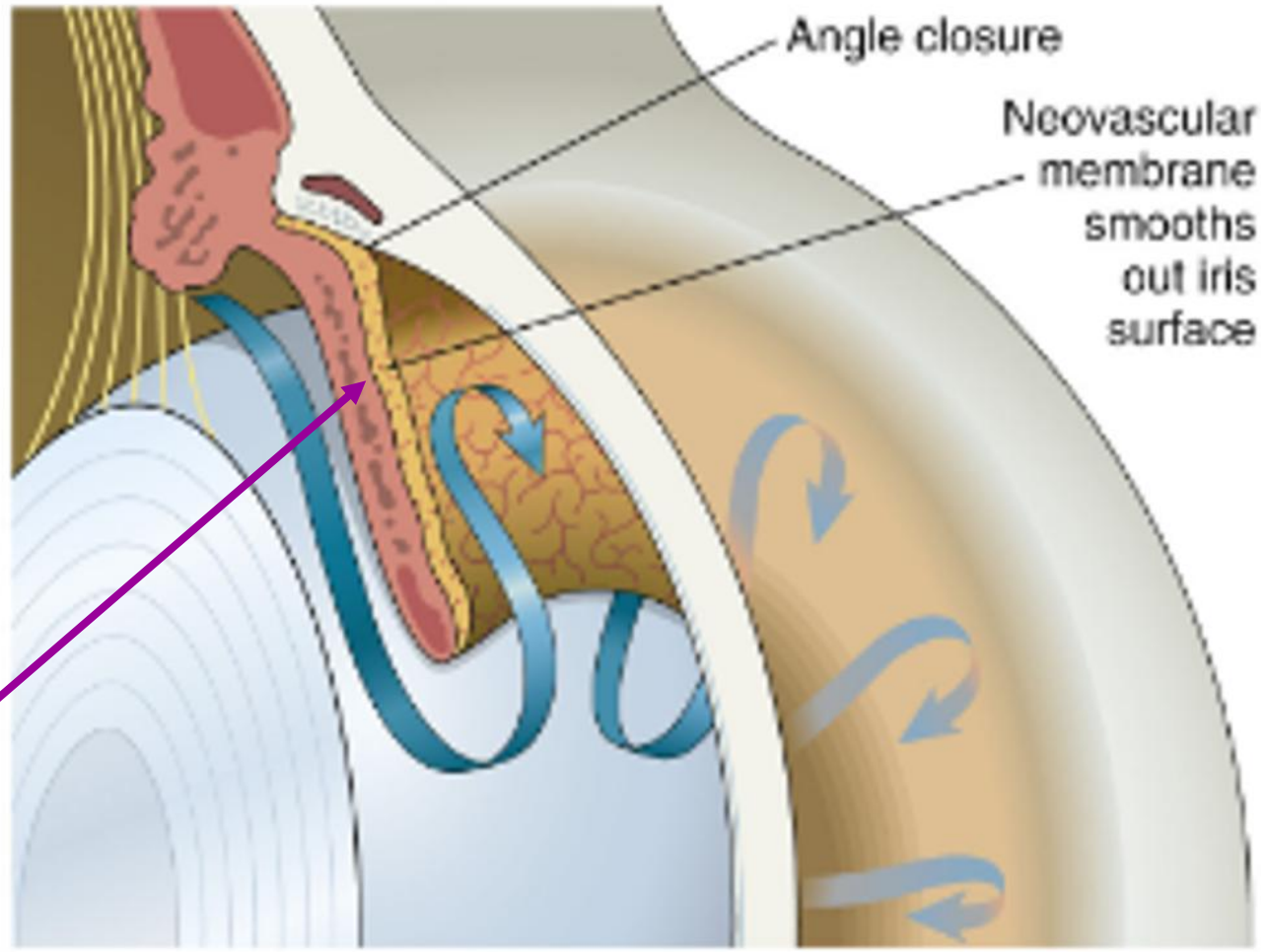


Figure 29.11

Glaucoma

- Cupping of the optic nerve
  - Seen in advanced case
- Other histologic changes
  - Diffuse loss of ganglion cells
  - Thinning of retinal nerve

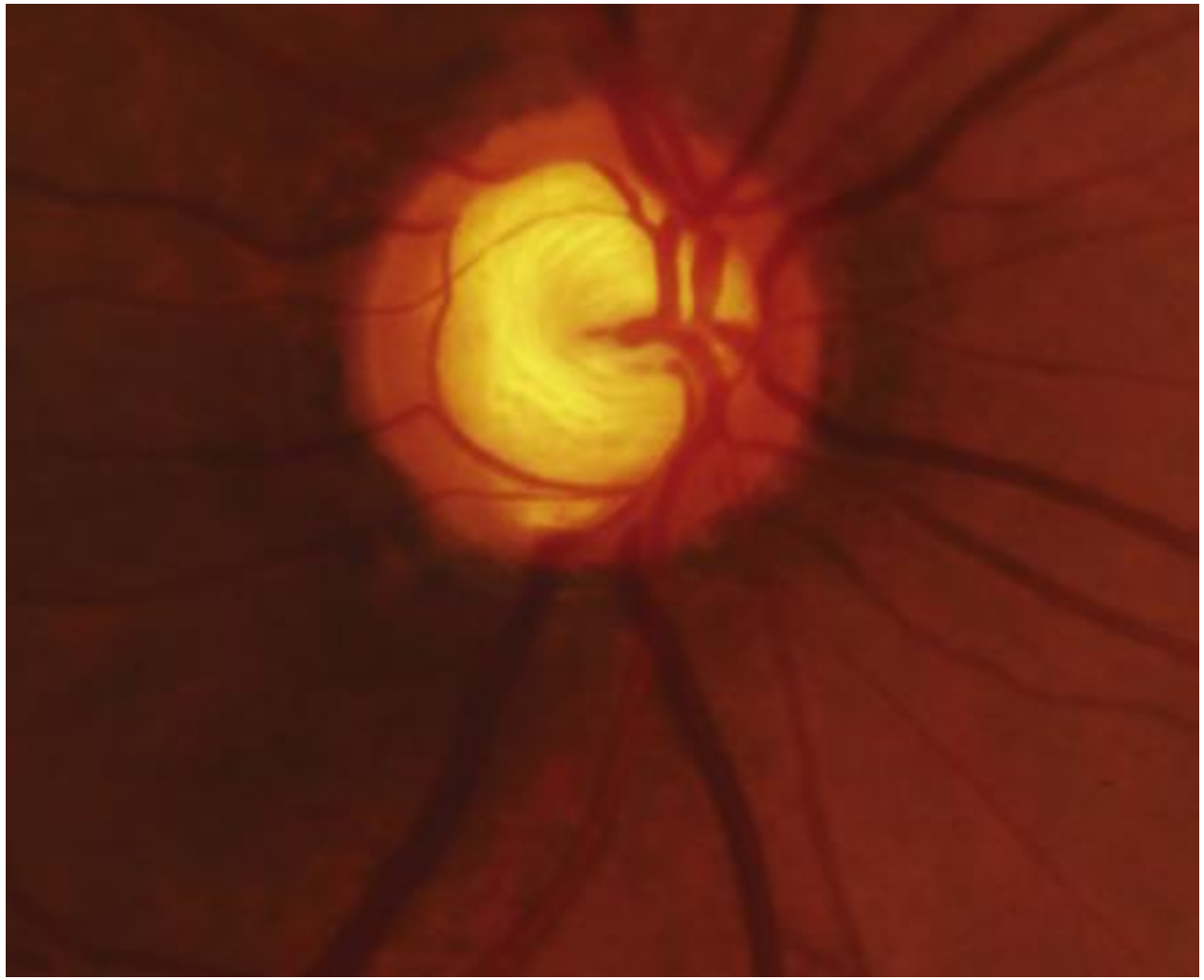


Figure 29.26

Glaucoma

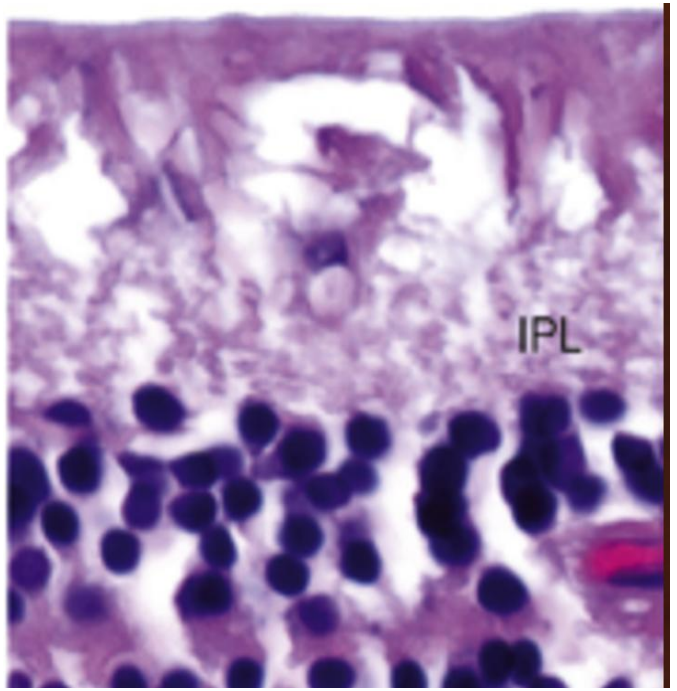
- Cupping of the optic nerve
  - Seen in advanced case
- Caused by prolonged increased IOP and subsequent atrophy



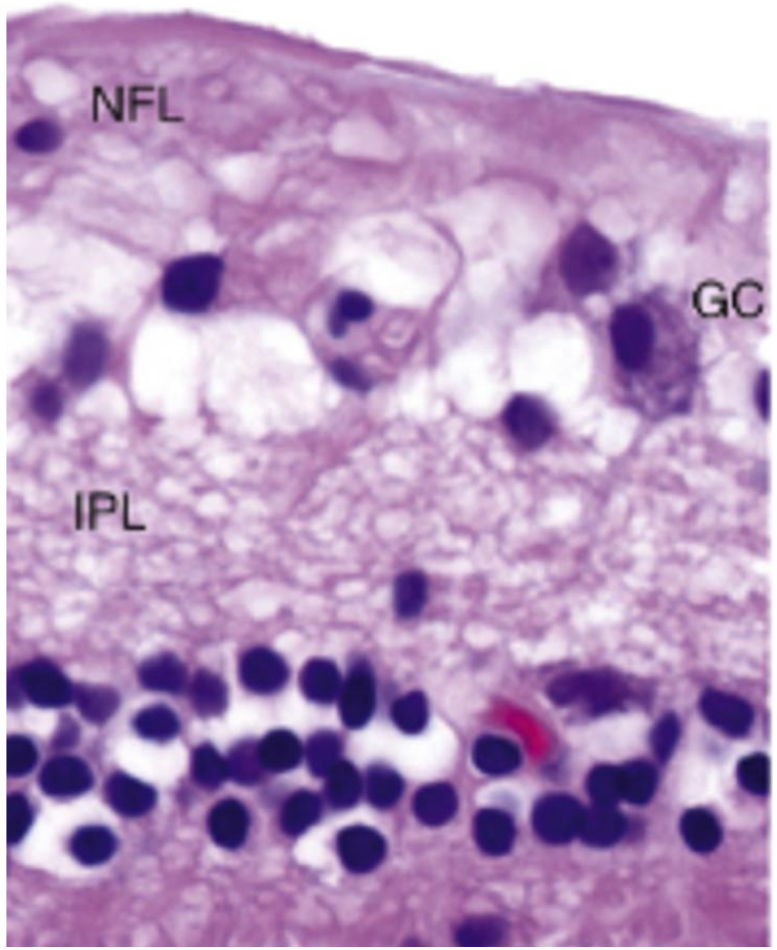
https://kellogg.umich.edu/theeyeshaveit/otherc/glaucoma.html

Glaucoma

- Histologic changes
  - Diffuse loss of ganglion cells



GLAUCOMA



NORMAL

Figure 29.26

Glaucoma

- Histologic changes
  - Thinning of retinal nerve
  - (large subdural space)

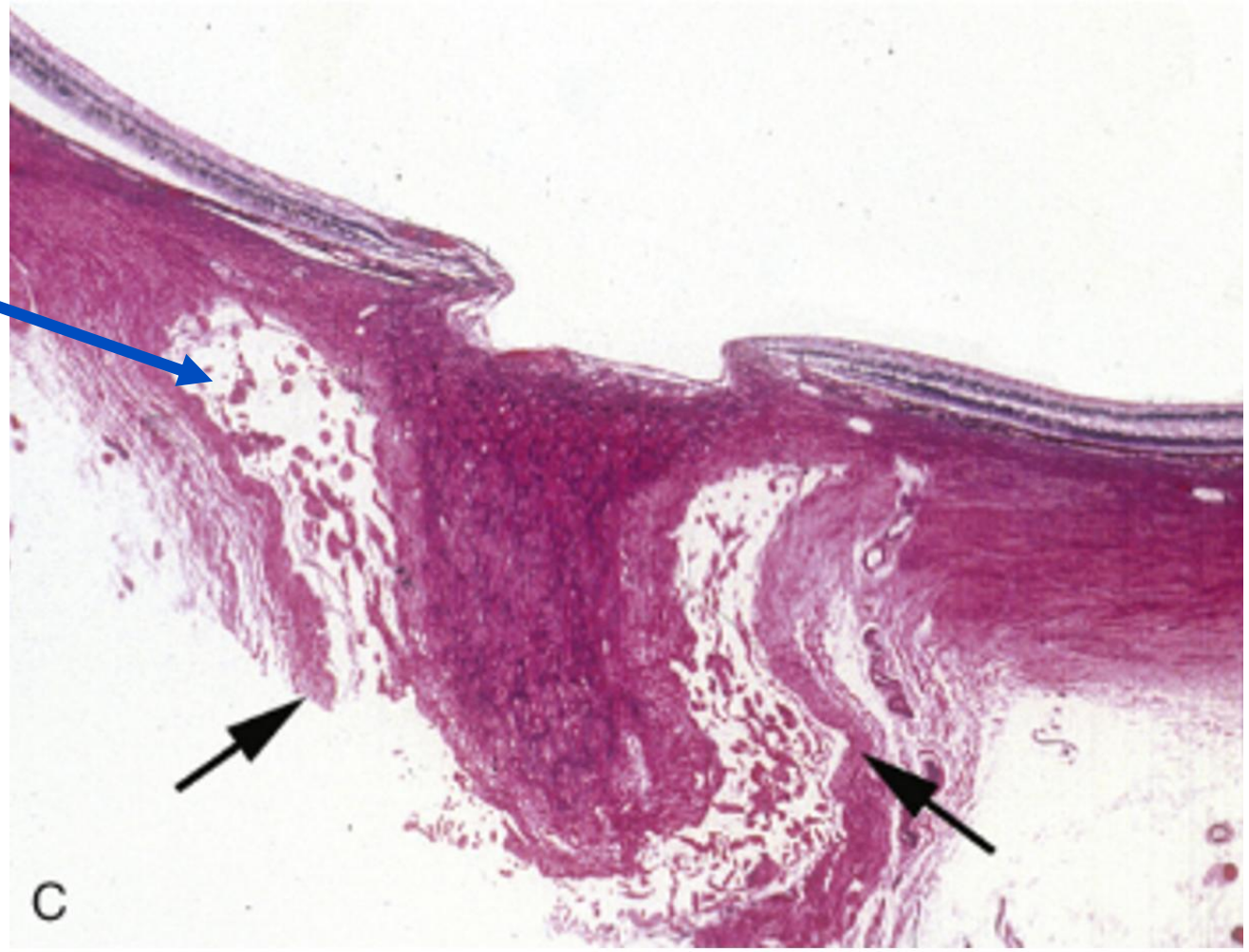


Figure 29.26

Glaucoma

- Diabetes mellitus
  - Iris
    - Neovascular membrane formation
      - Blocks trabecular network
  - Ciliary body (shown)
    - Ciliary body
    - Thickening of basement membrane
    - May decrease aqueous humor production

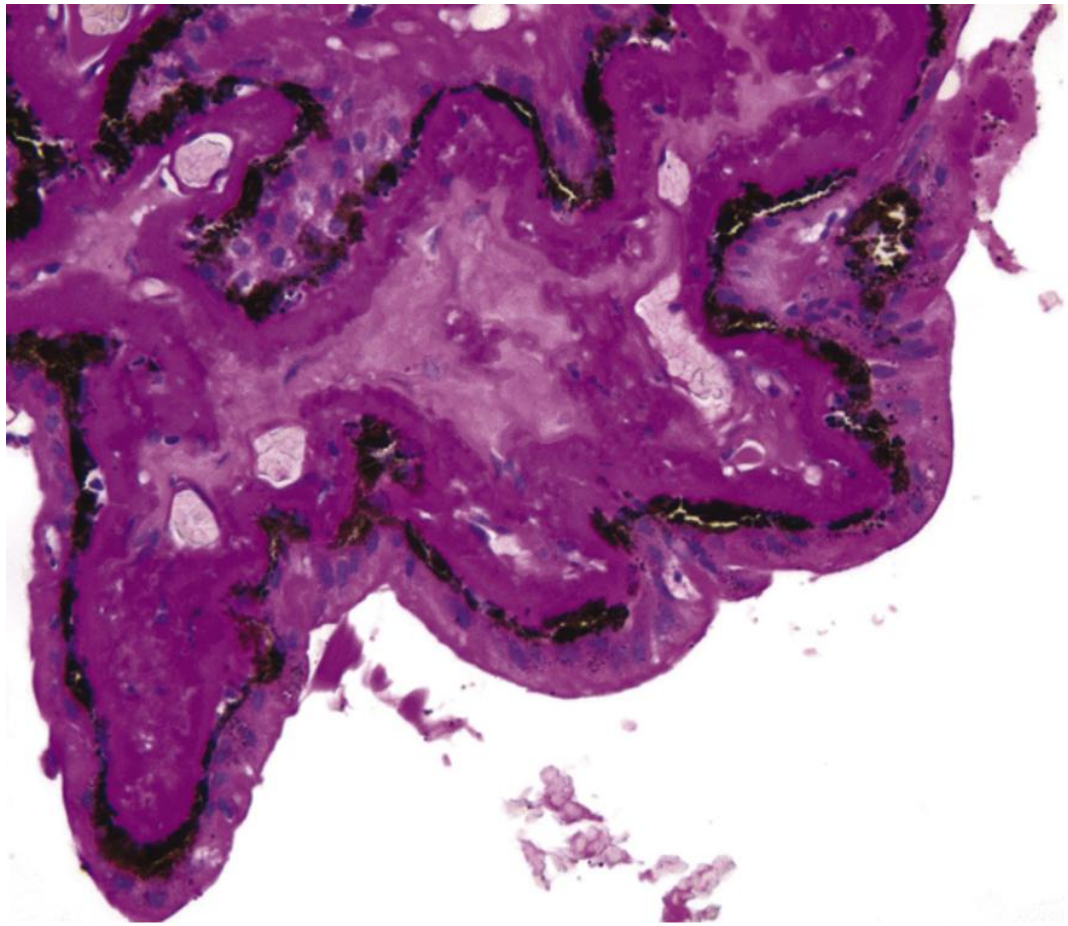


Figure 29.20

Autoimmune disorders of the eye

- Presentation
  - Typical clinical symptoms
    - Redness, pain, dryness, blurred vision, photophobia, and a foreign-body sensation
    - Symptoms can vary depending on the specific disorder
  - Typical Findings:
    - Inflammation of various parts of the eye, including the uvea (uveitis), cornea (keratitis), and sclera (scleritis).
- Morphology
  - Uvea
    - Inflammation of the uvea, leading to redness and pain
  - Cornea
    - Inflammation or ulceration of the cornea, causing pain and visual disturbances
  - Sclera
    - Thickening and inflammation of the sclera, resulting in redness and discomfort.

Uveitis

- Uvea consists of
  - Iris
  - Choroid
  - Ciliary body
- Uveitis causes
  - Autoimmune
    - Sympathetic ophthalmia
      - Condition that blinded Louis Braille
    - Juvenile idiopathic arthritis
  - Granulomatous
    - Caused by sarcoidosis

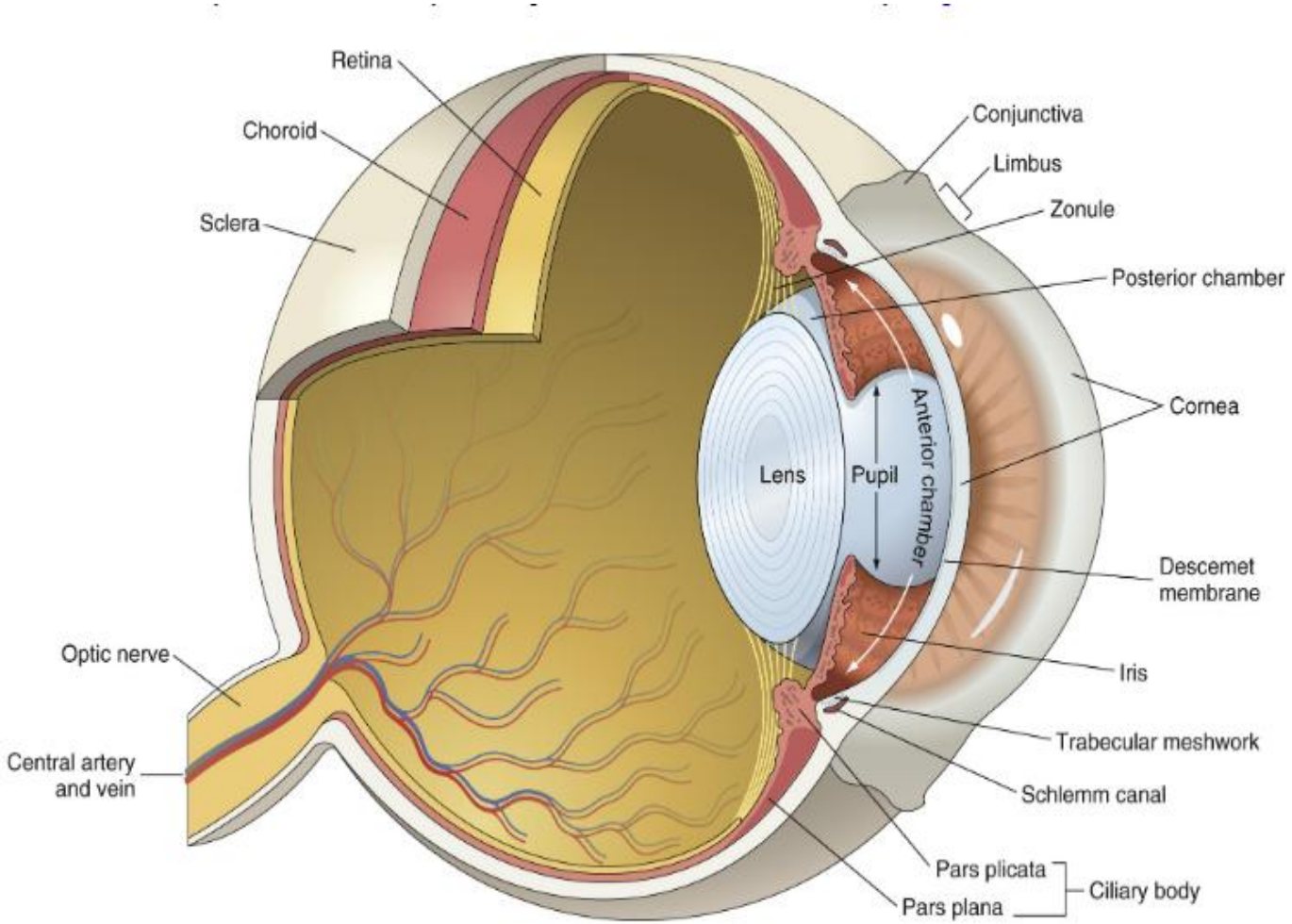
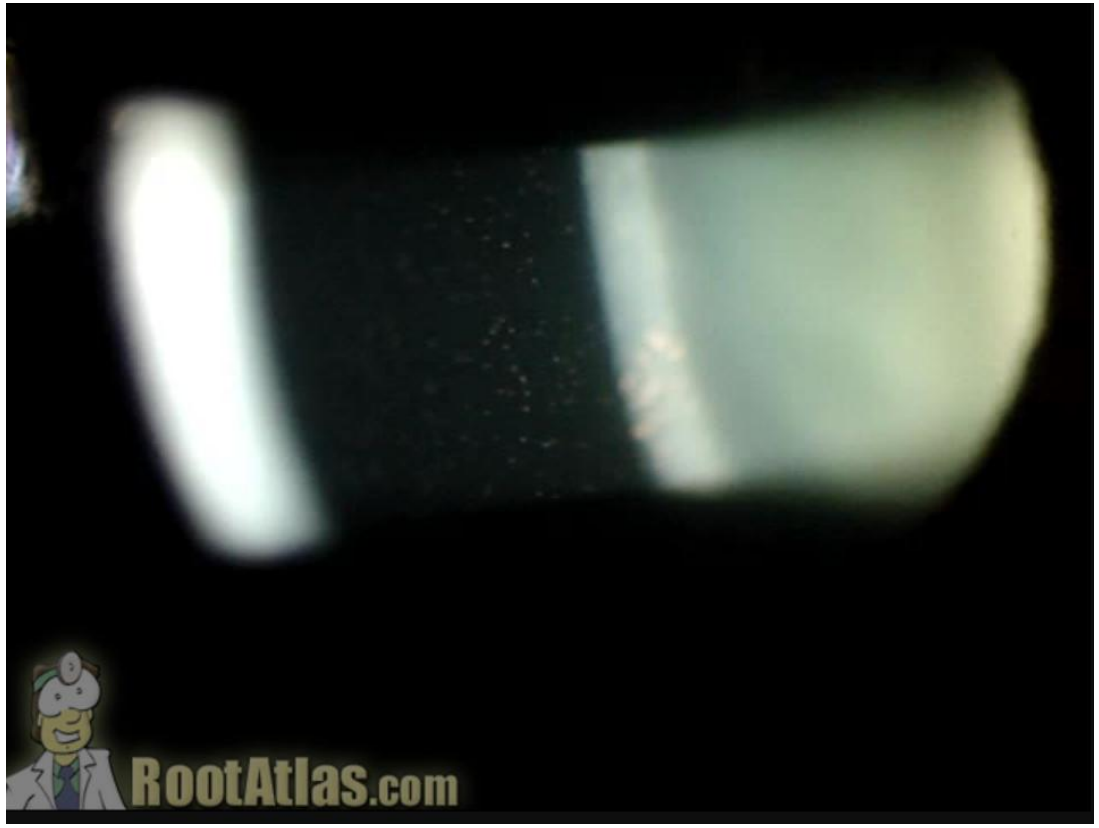


Figure 29.1

Uveitis

- Slit lamp examination
  - anterior chamber
- Cell and flare
  - Cells are white blood cells
  - Flare is hazy light diffusion from debris
  - seen in uveitis, iritis
    - Autoimmune
    - Other causes
      - Trauma
      - Infection
      - idiopathic



https://timroot.com/cell-and-flare-in-the-eye-video/

Age related macular degeneration (AMD)

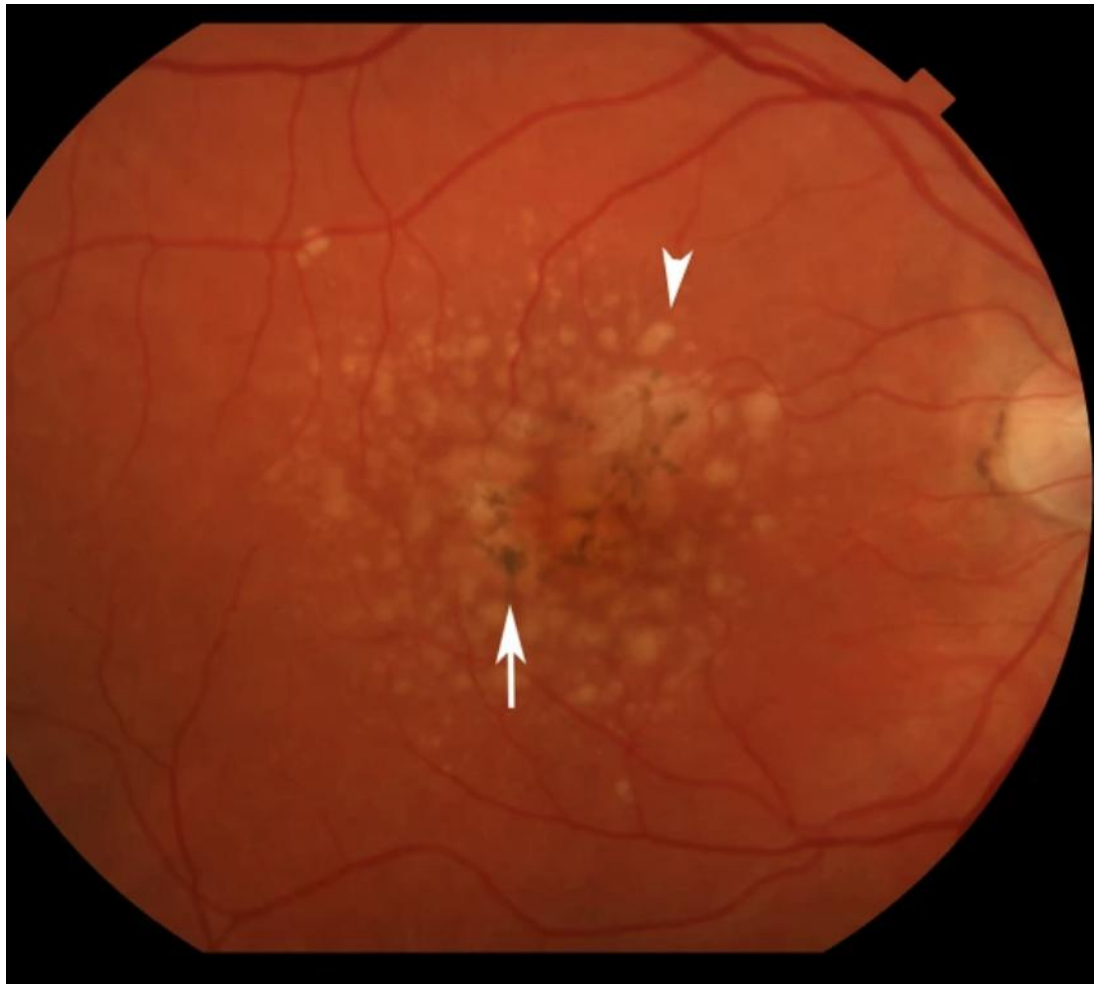
- Damage to the macula
  - Required for central vision
    - Needed for sharp, detailed, color rich central vision
  - About 8% of adults more than 75 years old
- Two forms
  - Dry (atrophic)
    - Loss of vision may be severe
    - Retinal drusen
  - Wet (neovascular)
    - VEGF mediated



https://kellogg.umich.edu/theeyeshaveit/opticfundus/retinal_drusen.html

Age related macular degeneration (AMD)

- Dry AMD.
  - Drusen (Arrowhead)
    - Yellowish/white deposits between retinal pigment epithelium (RPE) and Bruch's membrane
  - Hyperpigmentation (Arrow)
    - retinal pigment epithelium (RPE) debris
  - Thinning of the macula
  - Geographic atrophy of the macula
    - No longer round
    - Looks like "islands"



Handa, J.T., Bowes Rickman, C., Dick, A.D. *et al.* A systems biology approach towards understanding and treating non-neovascular age-related macular degeneration. *Nat Commun* **10**, 3347 (2019). https://doi.org/10.1038/s41467-019-11262-1

Age-related macular degeneration (AMD)

- Wet-type AMD
  - Neovascularization
    - Capillary-rich membrane forms between retinal pigment epithelium (RPE) and Bruch membrane (BM)
    - Tends to leak serum and bood leading to deposits and scars
    - Linked to increased VEGF
      - Anti-VEGF medications useful

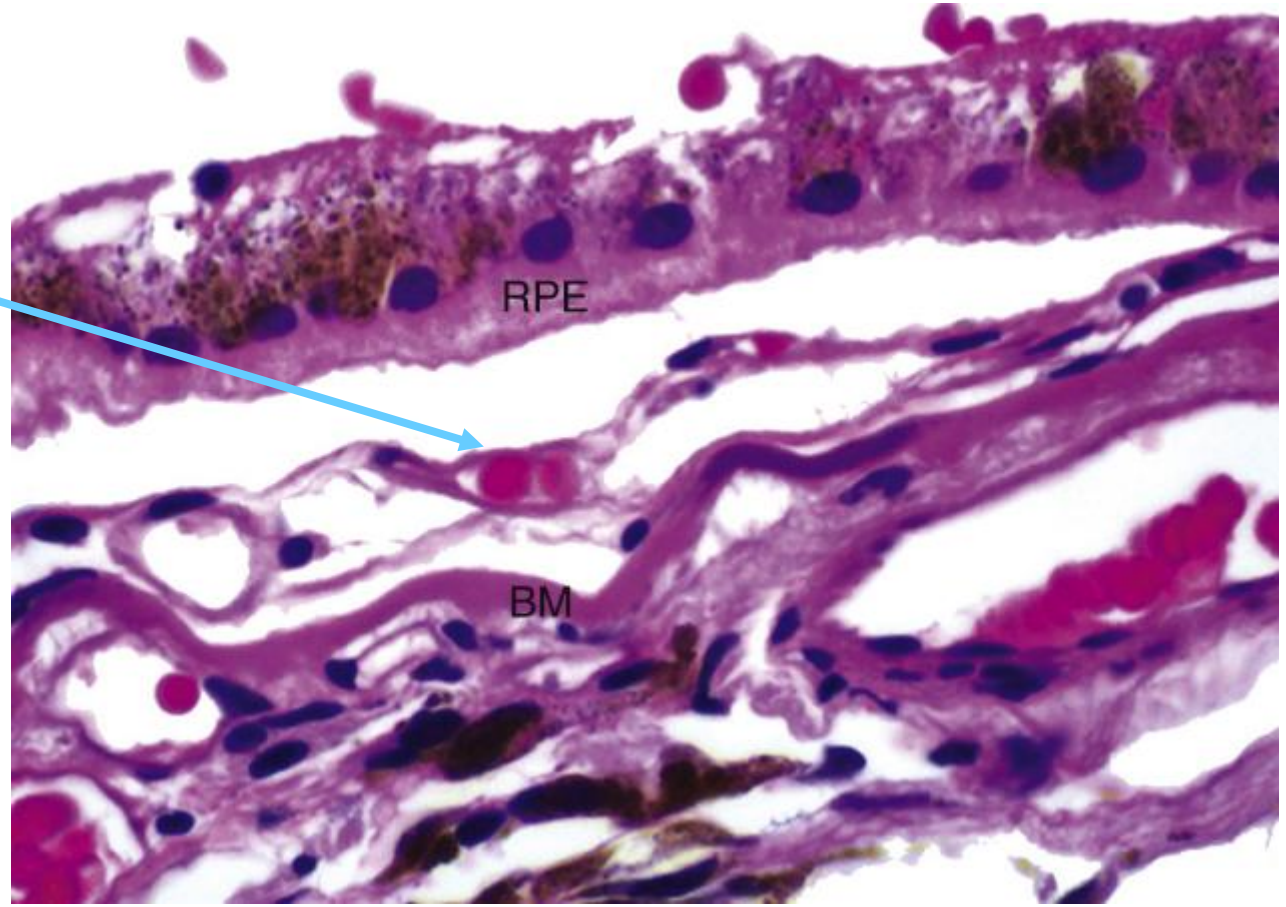


Figure 29.23

Age-related macular degeneration (AMD)

- Wet-type AMD
  - Neovascularization
  - Leakage of serum or blood into fovea



https://kellogg.umich.edu/theeyeshaveit/otherc/wet_macular_degeneration.html

Distinguishing Dry from Wet AMD

Dry AMD

- Progression
  - Slow and gradual
- Symptoms
  - Mild to moderate vision loss, difficulty reading, and needing more light for tasks
- Prognosis
  - Can progress to wet AMD
  - no definitive treatment but nutritional supplements may slow progression

Wet AMD

- Progression
  - Rapid and severe
- Pathology
- Symptoms
  - Sudden onset of visual distortions, dark spots, and rapid central vision loss
- Prognosis
  - More treatable with anti-VEGF injections, laser therapy, and photodynamic therapy
  - early detection is crucial

Uveal melanoma

- Clinical Presentation: Often asymptomatic and discovered incidentally during routine eye exams
  - Symptoms may include visual disturbances, blurred vision, floaters, flashes of light, and peripheral vision loss
  - Eye pain is rare but can occur in advanced cases
- Gross Morphologic Findings:
  - Appearance: Darkly pigmented mass, often brown or black
  - Location: Can occur in any part of the uvea (choroid, ciliary body, iris), but most commonly in the choroid
  - Size: Can vary, with larger tumors more likely to cause symptoms2

Uveal melanoma

- Histologic Findings:
  - Cell Type: Derived from melanocytes
  - Features: Spindle cell and epithelioid cell types, high mitotic rate, presence of necrosis, and vascular mimicry
  - Distinguishing Features: Poorly defined margins, rapid growth, and potential for extraocular extension
- Prognosis:
  - Uveal melanoma is the most common primary intraocular malignancy in adults, with a risk of metastasis to the liver in approximately 50% of cases
  - Risk of Metastasis: High, with liver being the most common site
- Monitoring
  - Regular follow-up exams and imaging to monitor for metastasis
- Management
  - Treatment options include enucleation, radiotherapy (e.g., proton beam irradiation), laser therapy, and systemic therapies for metastatic disease

Uveal melanoma

- Most common primary intraocular tumor of adults
  - 5% of melanomas
  - 5 year survival 80%
    - Much increasing risk afterwards
    - Liver metastasis common
- Gross photograph
  - Uveal melanoma arising from choroid
    - Darkly pigmented mass, often brown or black
    - Overlying retina detached

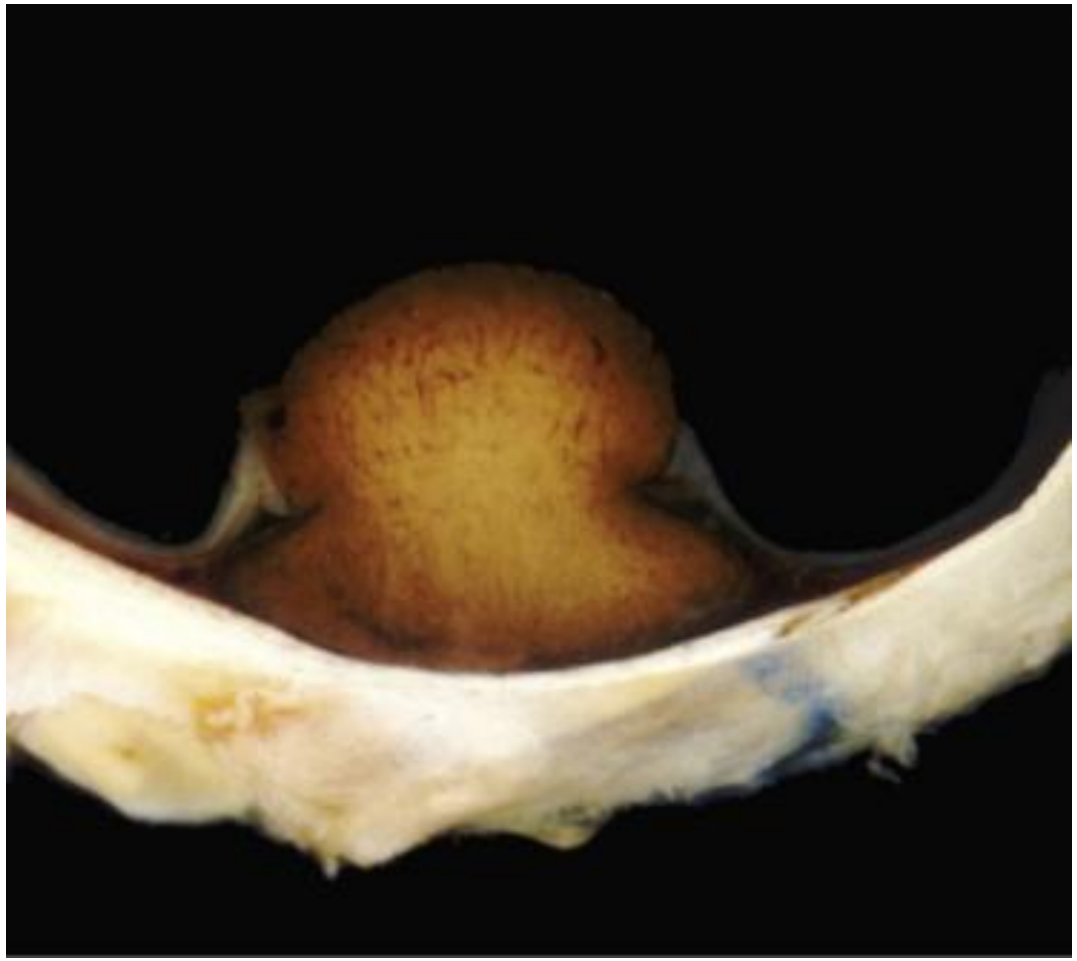


Figure 29.15

Uveal melanoma

- flat pigmented lesion of the choroid near the optic disc
- Genetics
  - Uveal melanomas do not contain BRAF mutations
  - Usually G-protein oncogene



Figure 29.15

Uveal melanoma

- Same patient several years later
- Tumor has grown
- ruptured through the Bruch membrane (between retina and choroid)



Figure 29.15

Uveal melanoma

- Histology
  - Epithelioid melanoma cells
    - Spherical
    - Nuclear atypia
    - Pleomorphism
  - Spindle cells (not shown)

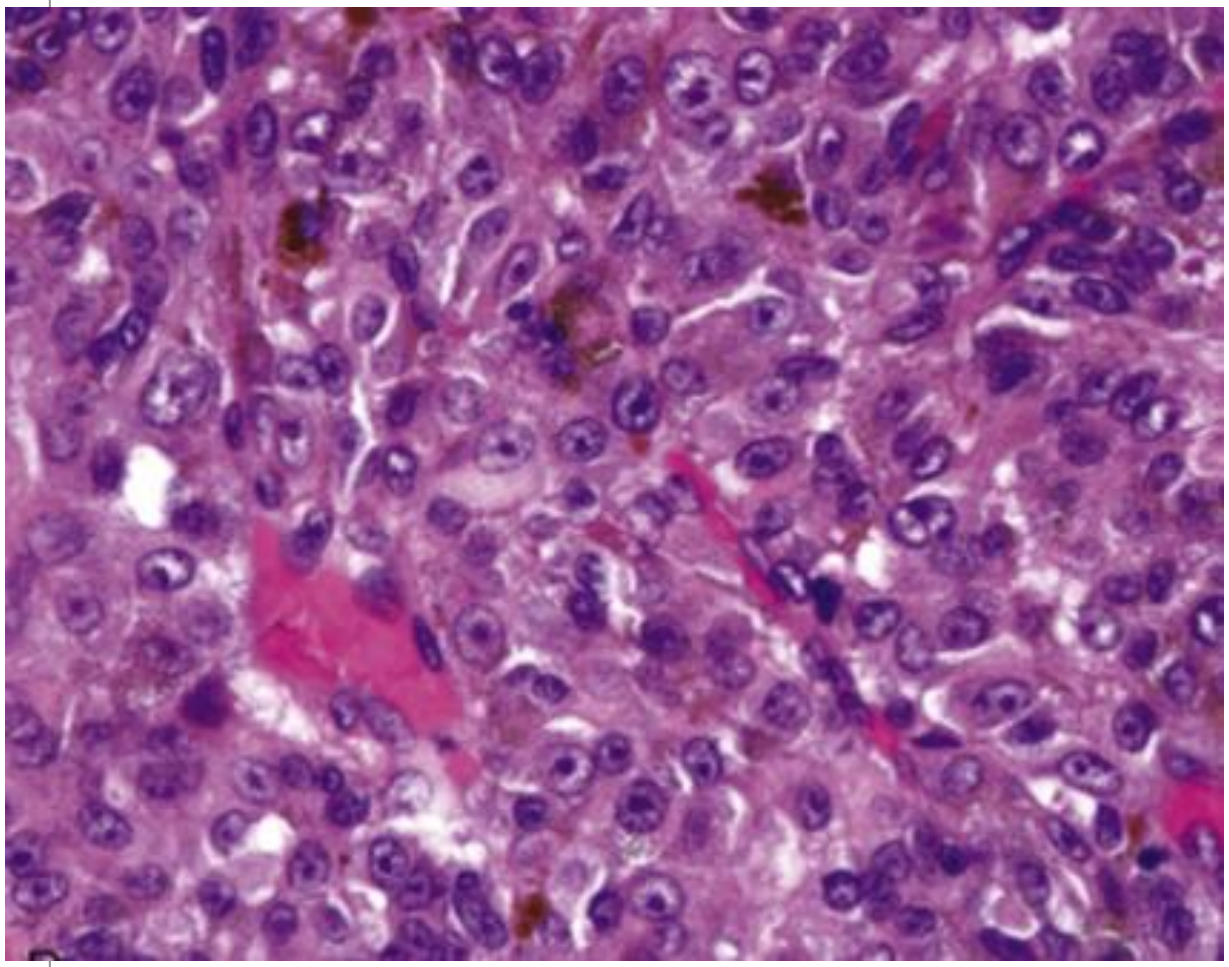


Figure 29.15

Posterior vitreous detachment

- Separation of the posterior section of the vitreous humor from the retina
- Older age greatest risk factor
- High risk for retinal detachment

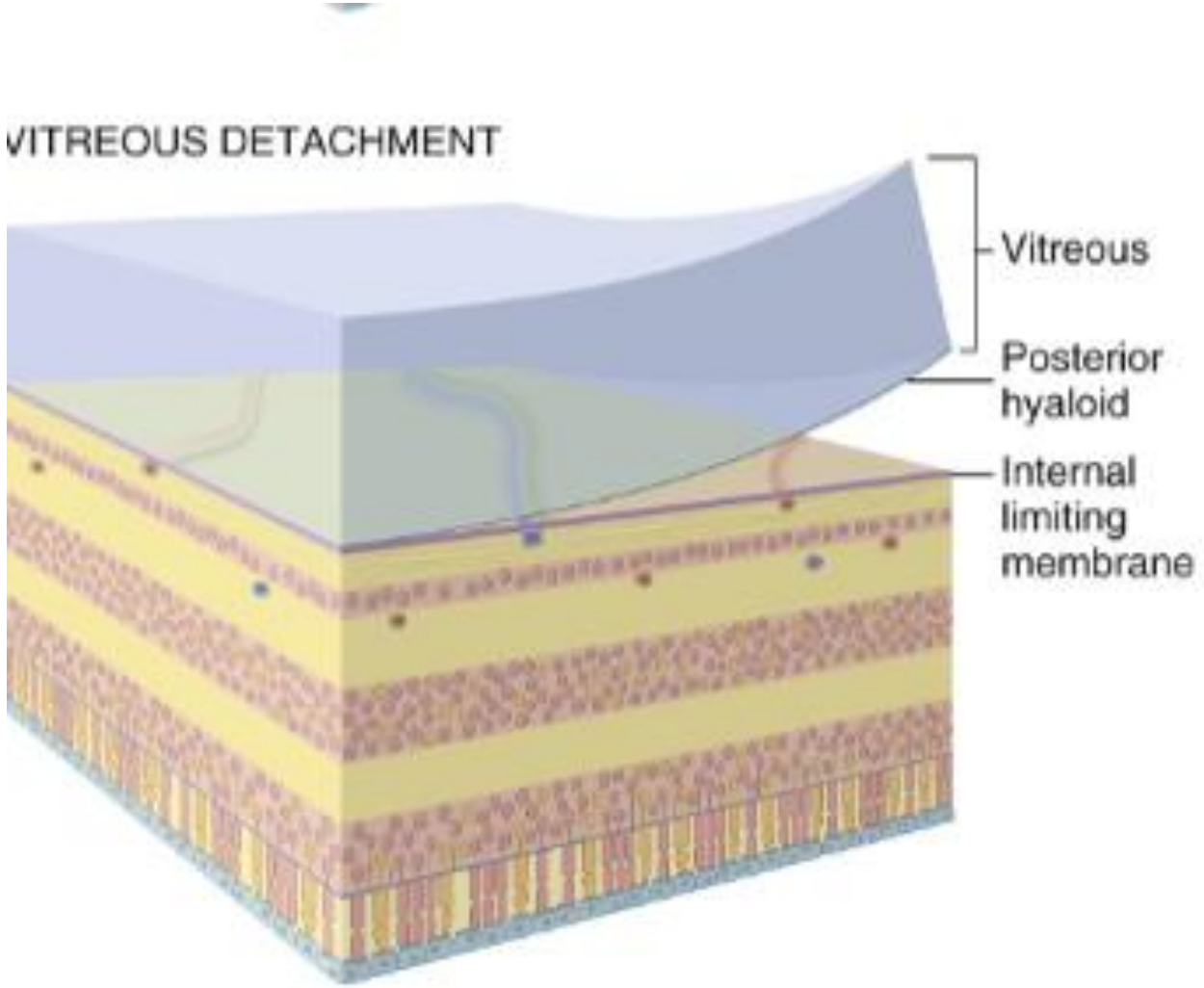
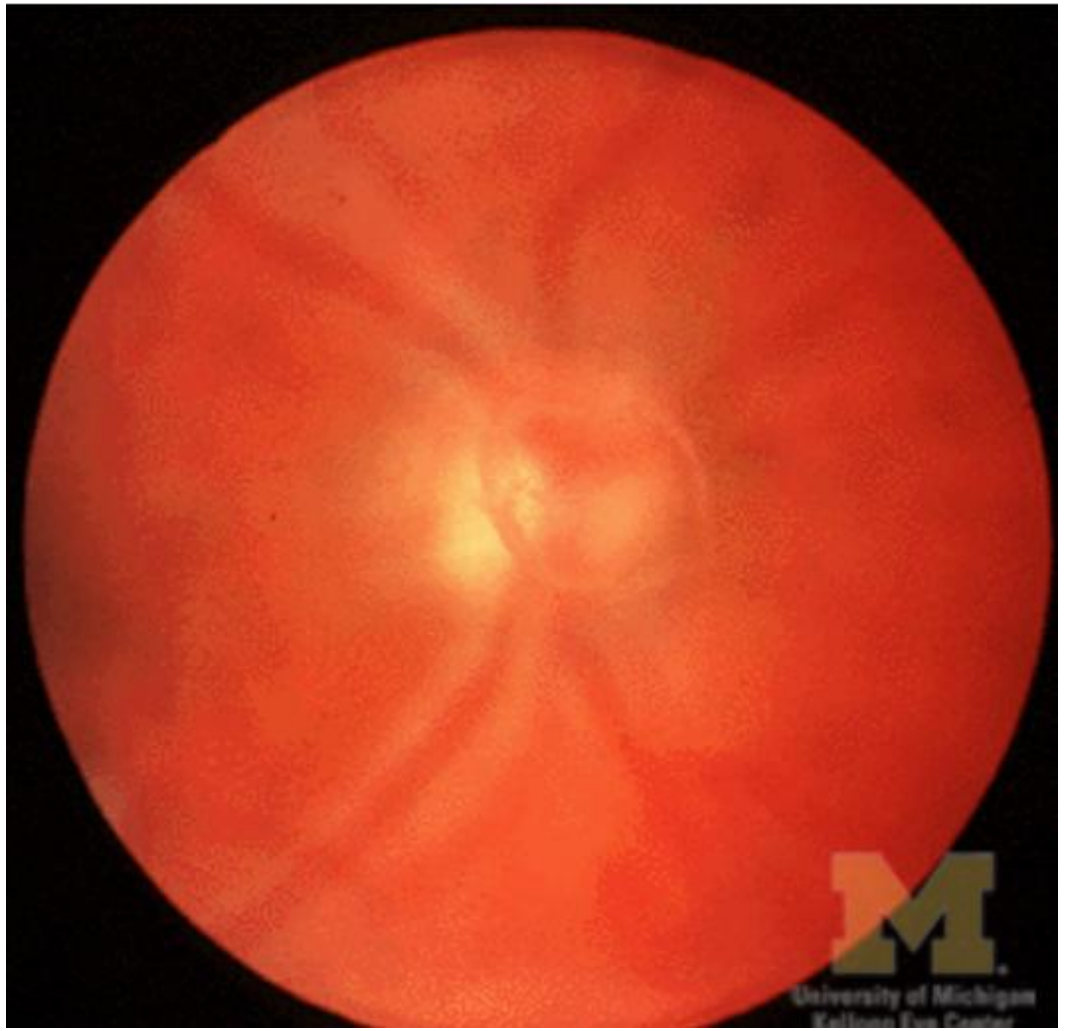


Figure 29.17

Posterior vitreous detachment

- Separation of the posterior section of the vitreous humor from the retina
  - Causes blurring and inability to discern features of the retina in detail
  - Somewhat related but not the same as floaters



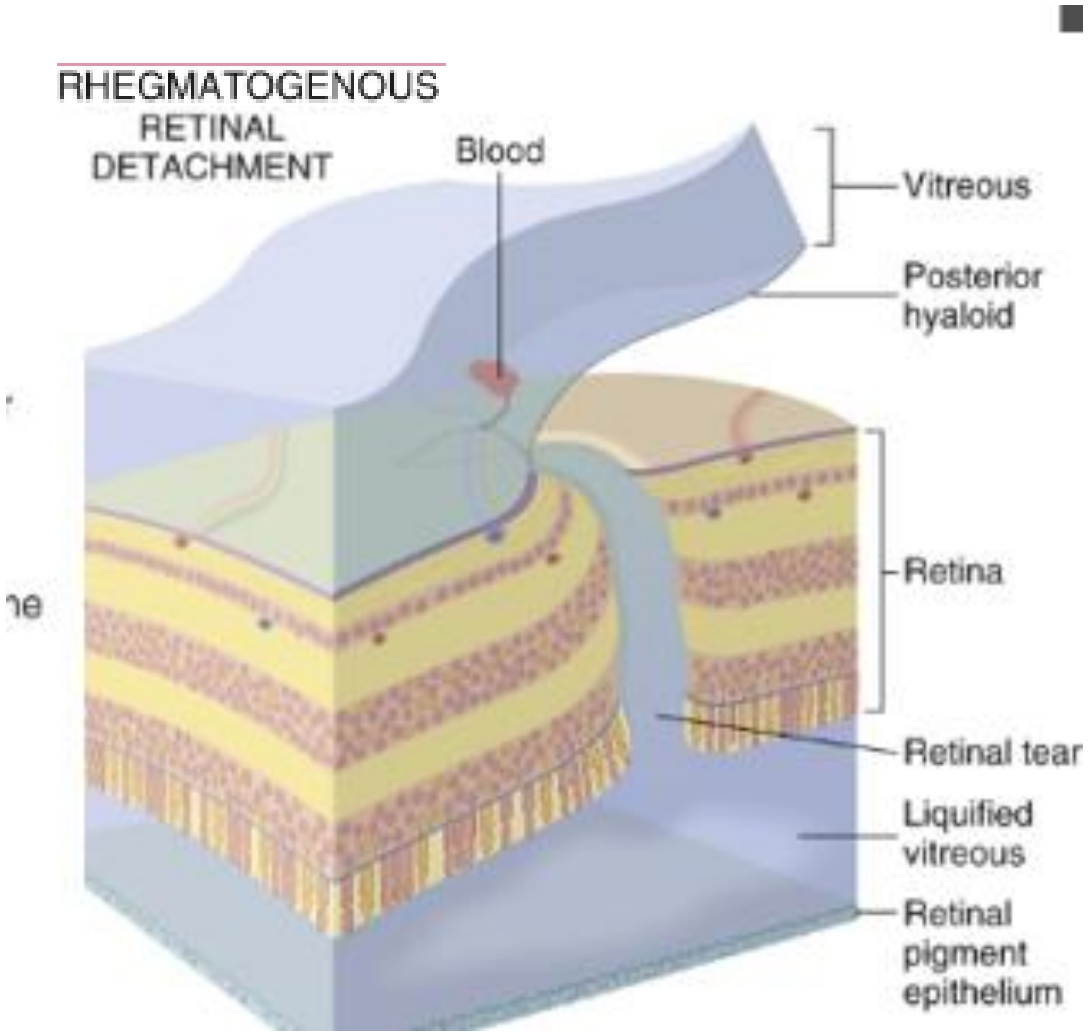
https://kellogg.umich.edu/theeyeshaveit/otherc/vitreous_detachment.html

Retinal detachment

- Presentation
  - Sudden appearance of floaters (dark spots or squiggly lines), flashes of light, and a shadow or curtain over the field of vision
  - Typical Findings
    - Reduced visual acuity, altered red reflex, and visual field defects
- Prognosis
  - General Outcome
    - If untreated, retinal detachment can lead to permanent vision loss
- Management
  - Prompt surgical intervention
  - Laser treatment, pneumatic retinopexy, or vitrectomy can reattach the retina and improve visual outcomes

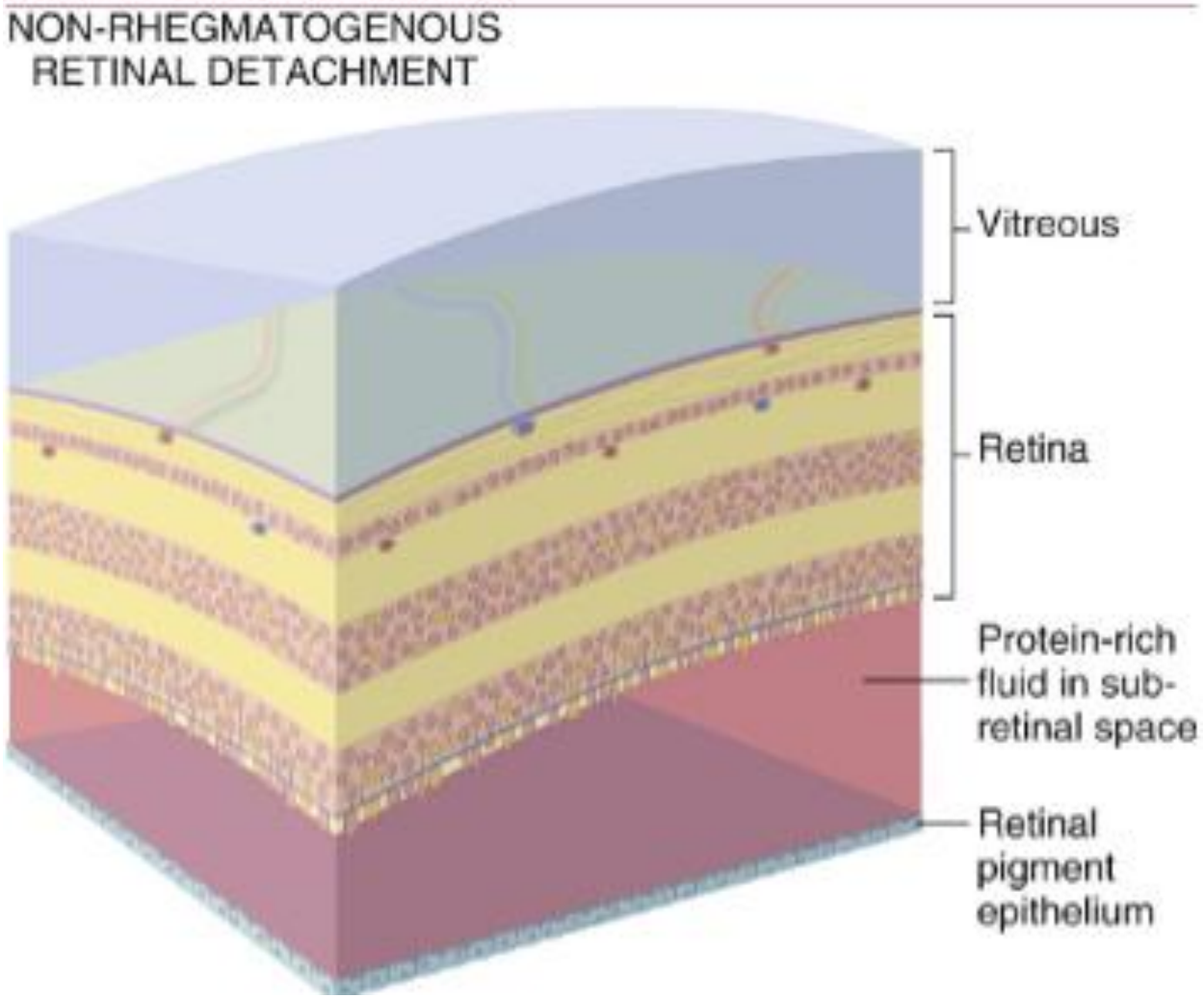
Retinal detachment

- Rhegmatogenous retinal detachment
  - Most common type
  - Liquified vitreous fluid floods between retina and RPE (retinal pigment epithelium)



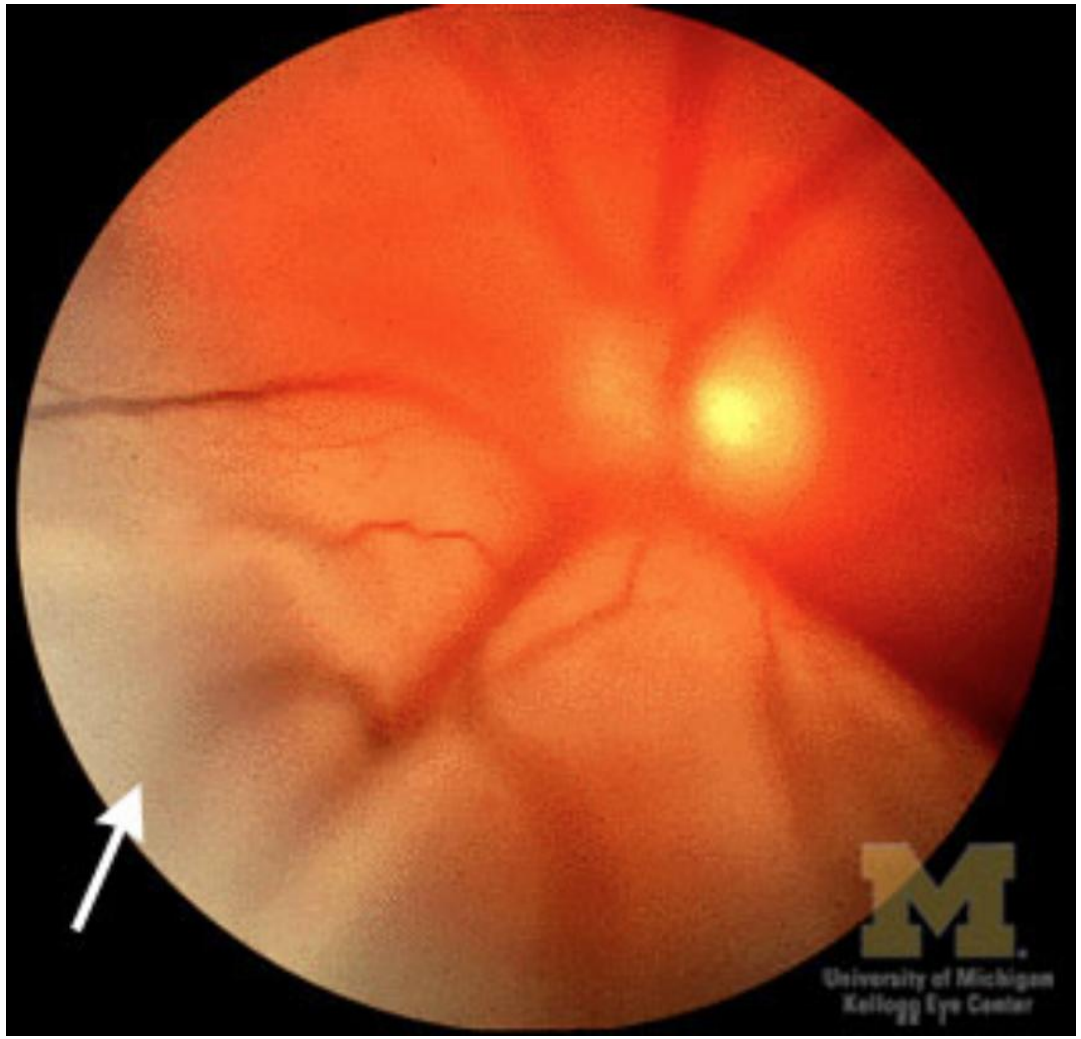
Retinal detachment

- Non-rhegmatogenous retinal detachment
  - Protein rich fluid accumulates in sub-retinal space
- Major risk factors
  - Choroidal tumors
  - Malignant hypertension



Retinal detachment

- Billowy folds of detached retina (arrow)
- Separation of the neurosensory retina from the underlying retinal pigment epithelium and choroid
- Histology (not shown)
  - Retinal Tissue
    - Degeneration of photoreceptors (rods and cones) when detached
  - Subretinal Space
    - Presence of fluid or blood in the sub-retinal space



https://kellogg.umich.edu/theeyeshaveit/otherc/retinal_detachment.html

Diabetic retinopathy

- Leading cause of blindness in American adults
  - Often asymptomatic in early stages; later stages may include blurred vision, floaters, dark areas in vision, difficulty reading, and poor night vision
- Diabetes forms many ocular lesions
  - Retinopathy
    - Most common pattern
  - Cataract formation
  - Glaucoma
- Diabetic retinopathy types
  - Nonproliferative diabetic retinopathy
  - Proliferative diabetic retinopathy

Diabetic retinopathy

- Nonproliferative diabetic retinopathy
  - Microangiopathy (Thickening of renal capillaries)
    - Thickened basement membrane
    - Pericyte loss
  - Retinal hemorrhages and exudates
    - Cotton wool spots
  - Microaneurysms
    - Saccular dilations of choroidal capillaries
    - Red dots on fundoscopic examination
  - Venous dilations
  - Diagnosis may include intravenous injection of due (fluorescein)

Diabetic retinopathy

- Cotton wool spots
  - Seen in nonproliferative diabetic retinopathy
  - Caused by hemorrhage and exudate in damaged vessels
  - Also seen in other systemic diseases that cause vascular damage
    - **Hypertension**
      - **More commonly seen with HTN than with DM**
    - Lupus



https://kellogg.umich.edu/theeyeshaveit/opticfundus/cotton_wool.html

Diabetic retinopathy

- Proliferative diabetic retinopathy
  - Retinal ischemia stimulates VEGF formation →
  - Retinal neovascularization
    - Newly formed vessels breach the retinal membrane (arrows)
      - Can also affect iris and cause glaucoma
      - Fragile – bleed extremely easily
    - Surface of optic nerve head or retina
    - Macular involvement → blindness

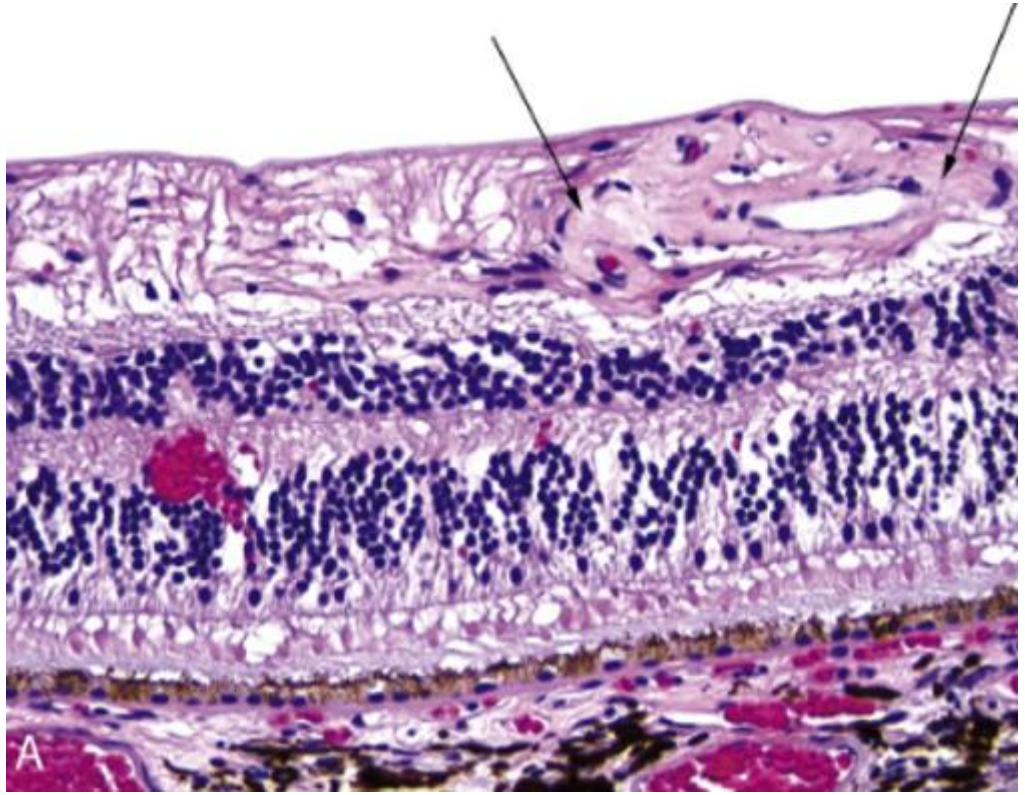


Figure 29.21

Diabetic retinopathy

- Proliferative diabetic retinopathy
  - Retinal neovascularization
    - Occurs after retinal ischemia, as a pathologic repair mechanism
  - Vitreous detachment can cause large bleeds
    - Pulls up on delicate tangle of vessels
  - VEGF helps mediate this process
    - Anti-VEGF medications are useful

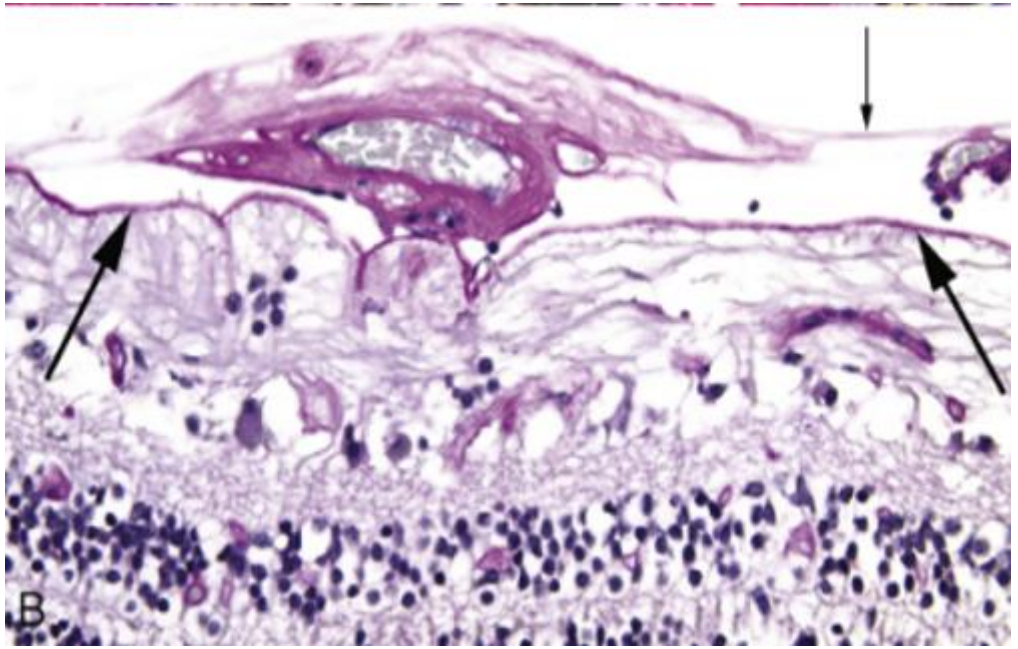
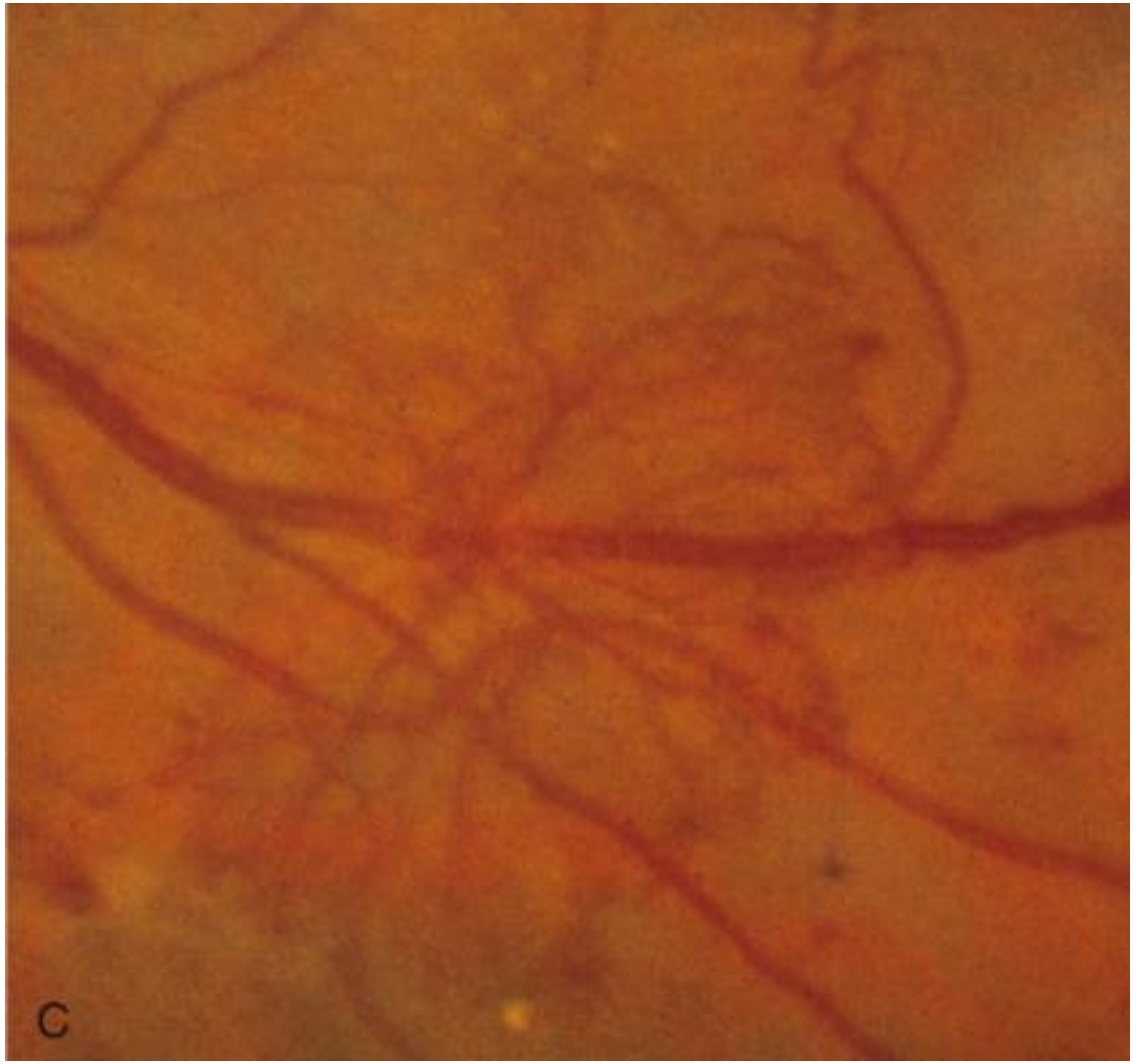


Figure 29.21

Diabetic retinopathy

- Proliferative diabetic retinopathy
  - Retinal neovascularization
- AKA neovascularization “elsewhere”
  - These tangles do not just involve optic disc



Hypertensive retinopathy

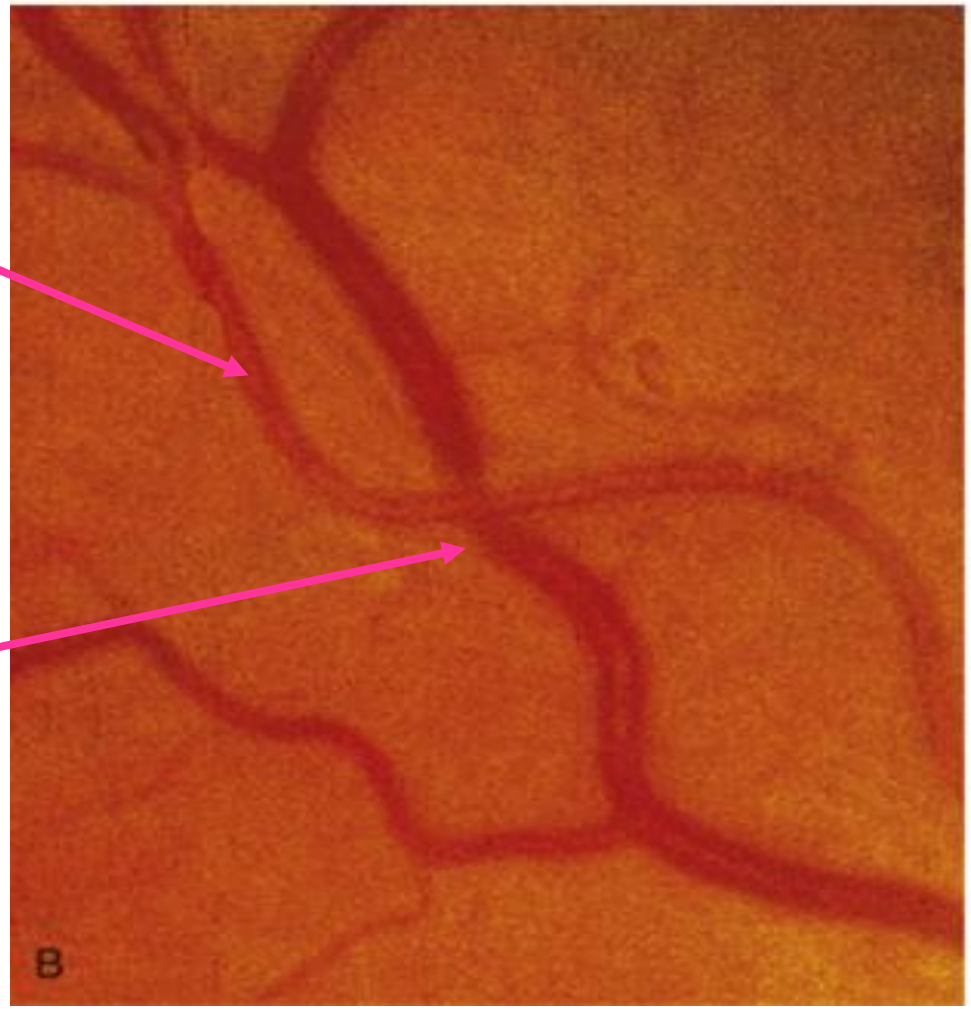
- Sclerotic changes in arterioles
  - Basement membrane thickening
    - Decreases width of column of blood
  - “Copper-wire”
  - “Silver-wire”



https://kellogg.umich.edu/theeyeshaveit/opticfundus/chronic_hypertension.html

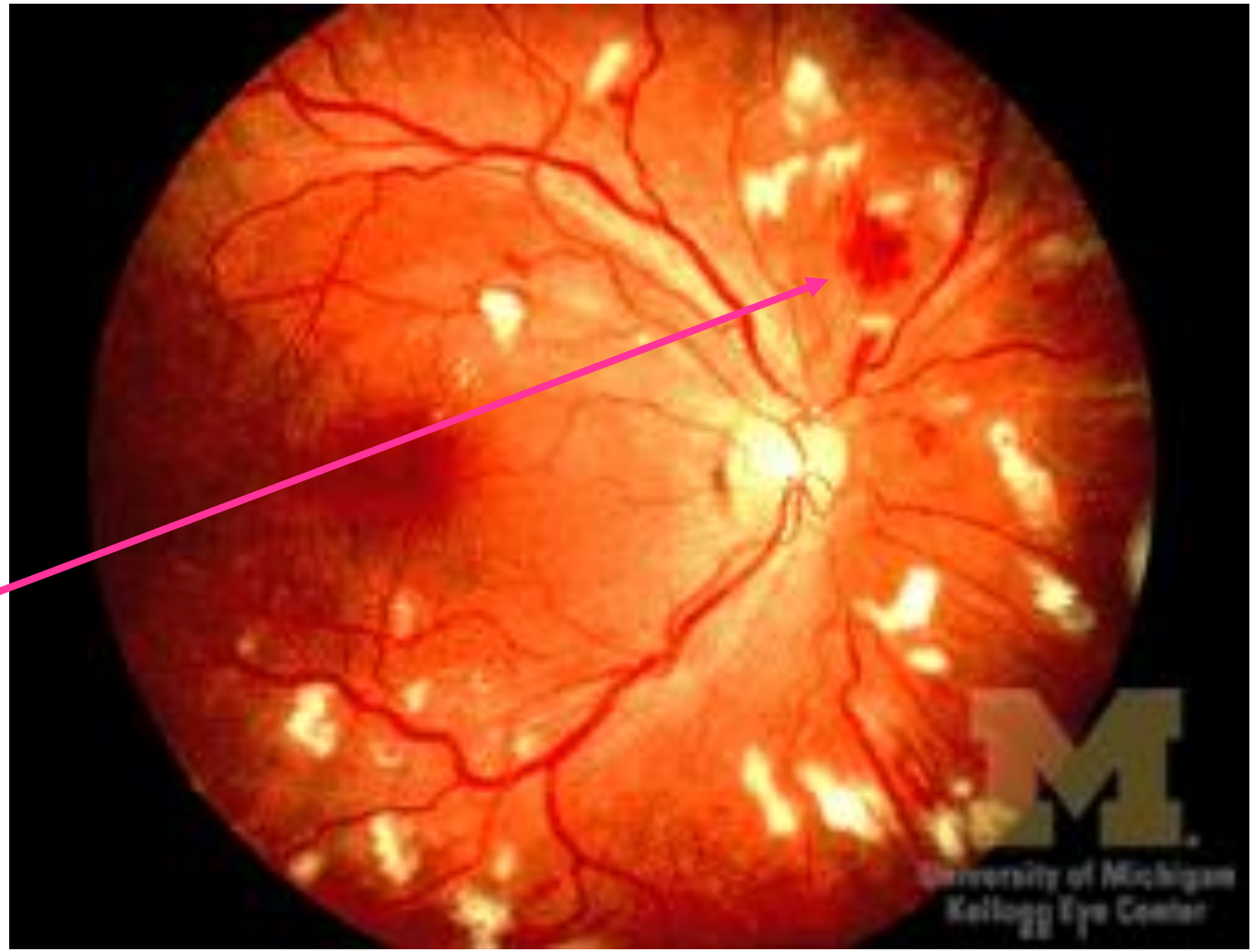
Hypertensive retinopathy

- Sclerotic changes in arterioles
  - Diameter of arteriole reduced
  - Basement membrane thickening
    - Decreases width of column of blood
- Arteriovenous (AV) Nicking
  - Underlying venule compressed
  - Vein is compressed where the sclerotic arteriole crosses over it



Hypertensive retinopathy

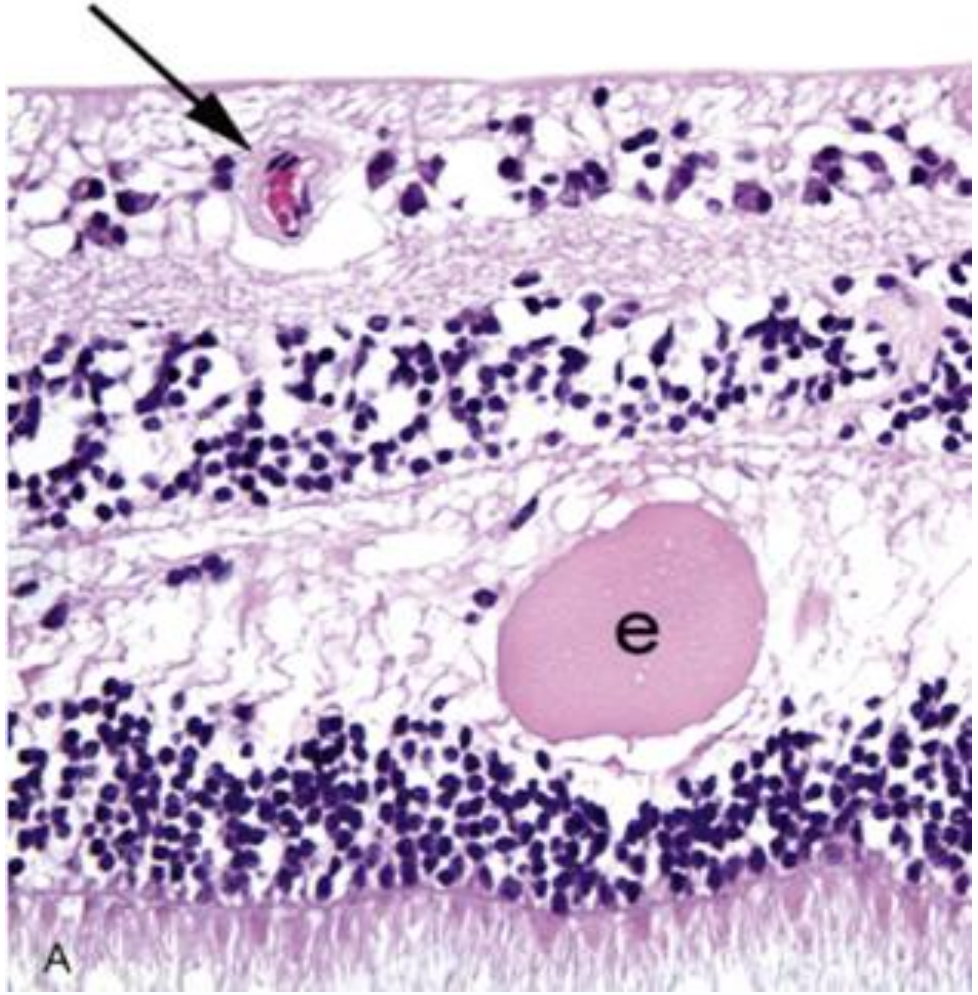
- Cotton wool spots
  - Infarcts of the nerve fiber layer of retina (cytoid bodies)
  - Also seen in other systemic diseases that cause vascular damage (mechanism may be different)
    - **Diabetes**
    - Lupus
- Retinal hemorrhages also seen



https://kellogg.umich.edu/theeyeshaveit/opticfundus/cotton_wool.html

Hypertensive retinopathy

- Sclerotic changes in arterioles (arrow)
  - Diameter of arteriole reduced
  - Basement membrane thickening
    - Decreases width of column of blood
    - “Copper-wire”
    - “Silver-wire”
- Exudate (e) forms cotton-wool spots



Hypertensive retinopathy

- Presentation
  - Often asymptomatic in early stages
    - advanced stages may present with vision loss, headaches, and visual disturbance
- Prognosis:
  - Early stages may be managed with blood pressure control
  - Advanced stages can lead to irreversible vision loss if not treated
- Malignant hypertension
  - May cause increased intracranial pressure → **papilledema**

Papilledema

- Caused by increased intracranial pressure
  - Medical emergency
- Findings
  - Swollen optic nerve
    - Indistinct edges do to swelling
  - Hyperemic optic nerve
    - Engorged with blood



Figure 29.25

Papilledema

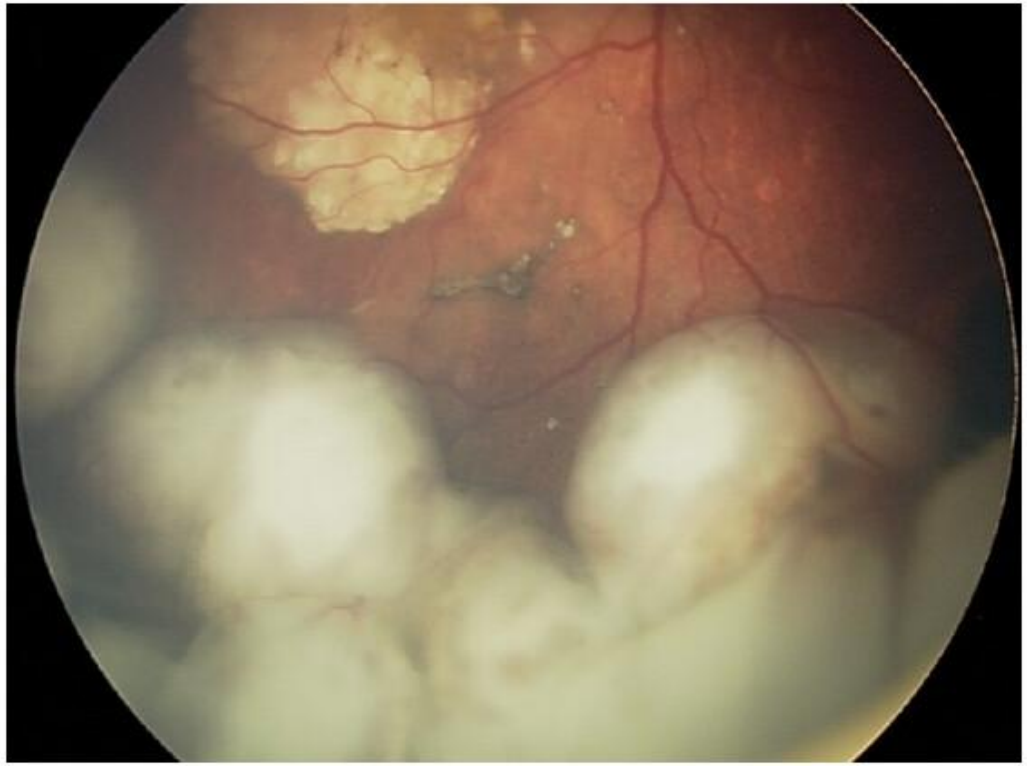
- Non-hypertensive causes (most common)
  - Causes
    - Idiopathic Intracranial Hypertension
    - Brain tumor or cyst
    - Stroke
  - Often treated with mannitol
- Malignant hypertension caused
  - Treatment
    - Nicardipine (fast acting calcium channel blocker)
    - Labetalol (fast acting beta blocker)



Figure 29.25

Retinoblastoma

- Leukocoria
  - white reflection from the retina – often in photos
    - Can be yellowish-white or off-white
  - Sometimes called “white reflex”
- strabismus
  - (misalignment of the eye)
- Poor vision in affected eyes
- High cure rate if intraocular



Zanaty, Mario, Barros, Guilherme, Chalouhi, Nohra, Starke, Robert M., Manasseh, Philip, Tjoumakaris, Stavropoula I., Shields, Carol L., Hasan, David, Bulsara, Ketan, Rosenwasser, Robert H., Jabbour, Pascal, Update on Intra-Arterial Chemotherapy for Retinoblastoma, *The Scientific World Journal*, 2014, 869604, 6 pages, 2014.

Retinoblastoma

- Presents in young children
  - Usually less than 5 years old
  - Older children may have slow growing, missed tumor
- Morphology
  - White, cream-like mass
    - Endophytic (grows into the vitreous)
    - Exophytic (grows into subretinal space)
  - Can fill entire globe

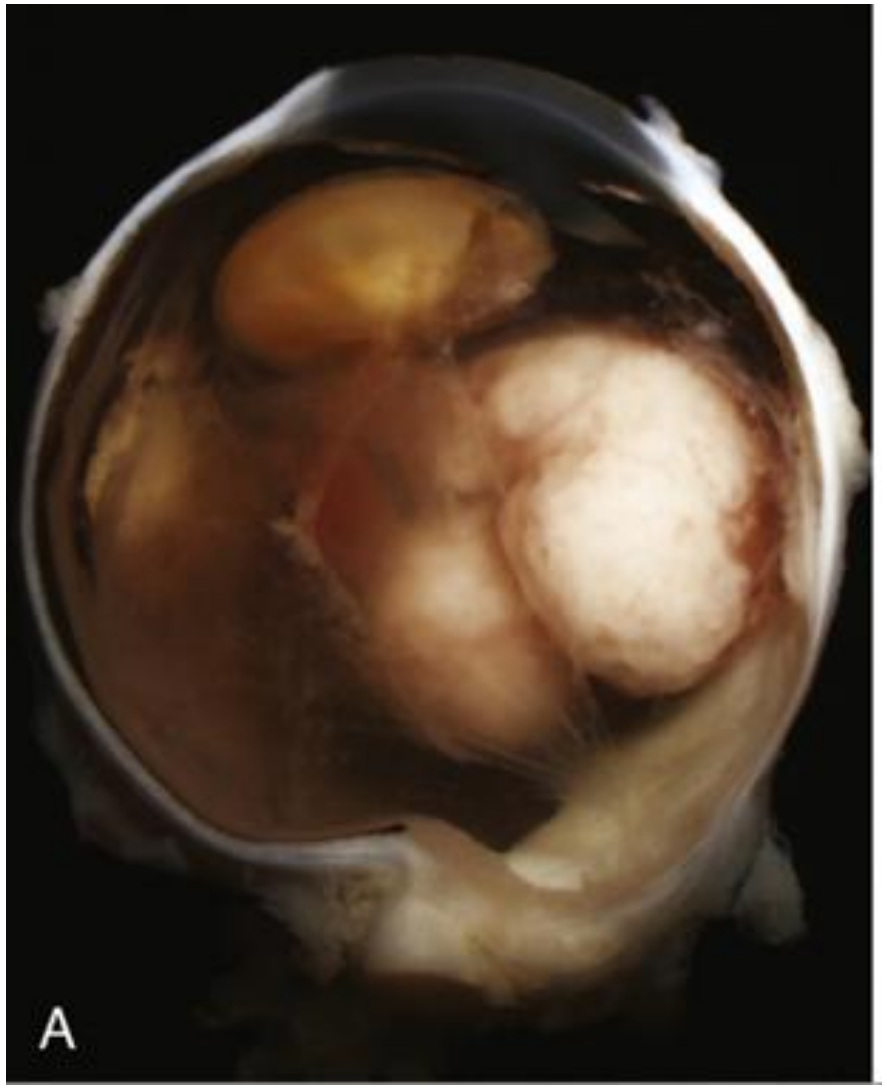


Figure 29.24

REVIEW

Retinoblastoma

- Tumor suppressor: RB The Governor of Proliferation
  - Prevents out of control DNA transcription
  - **Sporadic form**
    - **Effects one eye** with no additional cancer risk
    - 60% of case
  - **Familial form**
    - **10,000 times risk of developing RB (both eyes) and other cancers**
    - 40% of retinoblastoma is familial autosomal dominant

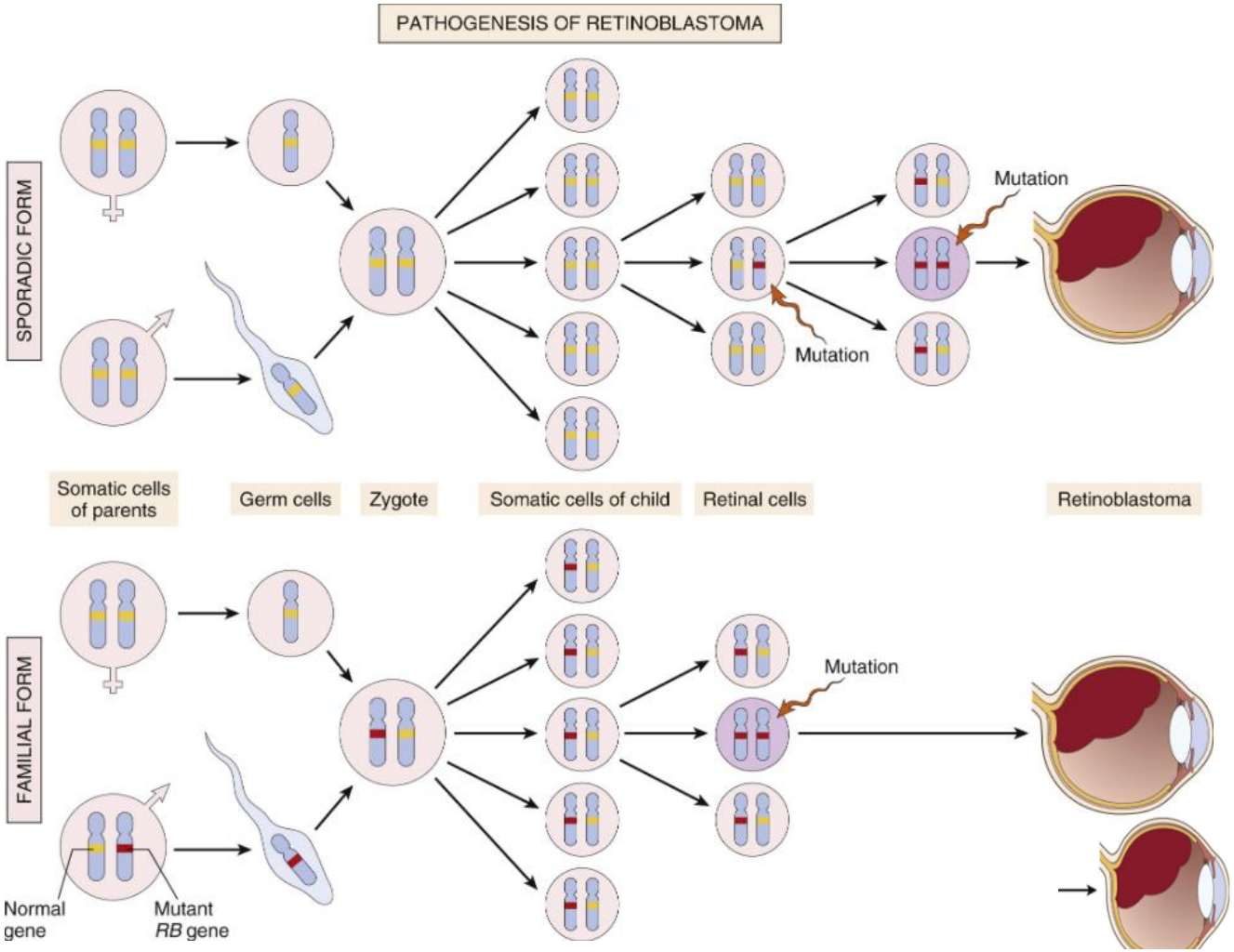


Figure 7.25

Retinoblastoma

- Small, round blue cells
- Flexner-Wintersteiner rosettes
  - (white arrow)
  - Arrangements of a single layer of neoplastic cells around a pseudolumen
- Dystrophic calcification
  - (black arrow)
- Tumor cells surround blood vessels

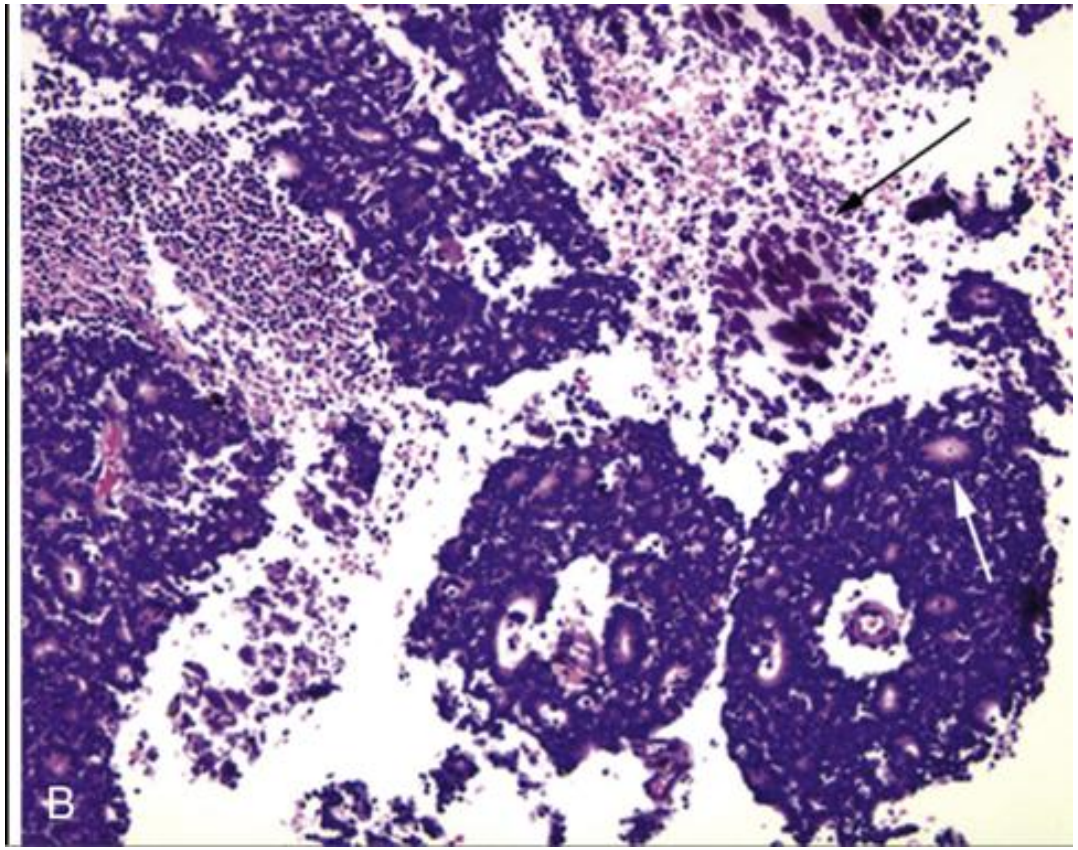


Figure 29.24

Cherry-red spot

- Storage diseases
  - Tay-sachs disease
  - Nieman-Pick disease
  - Excess storage material causes discoloration
- Note:
  - Similar lesion can be seen with total occlusion of central retinal artery

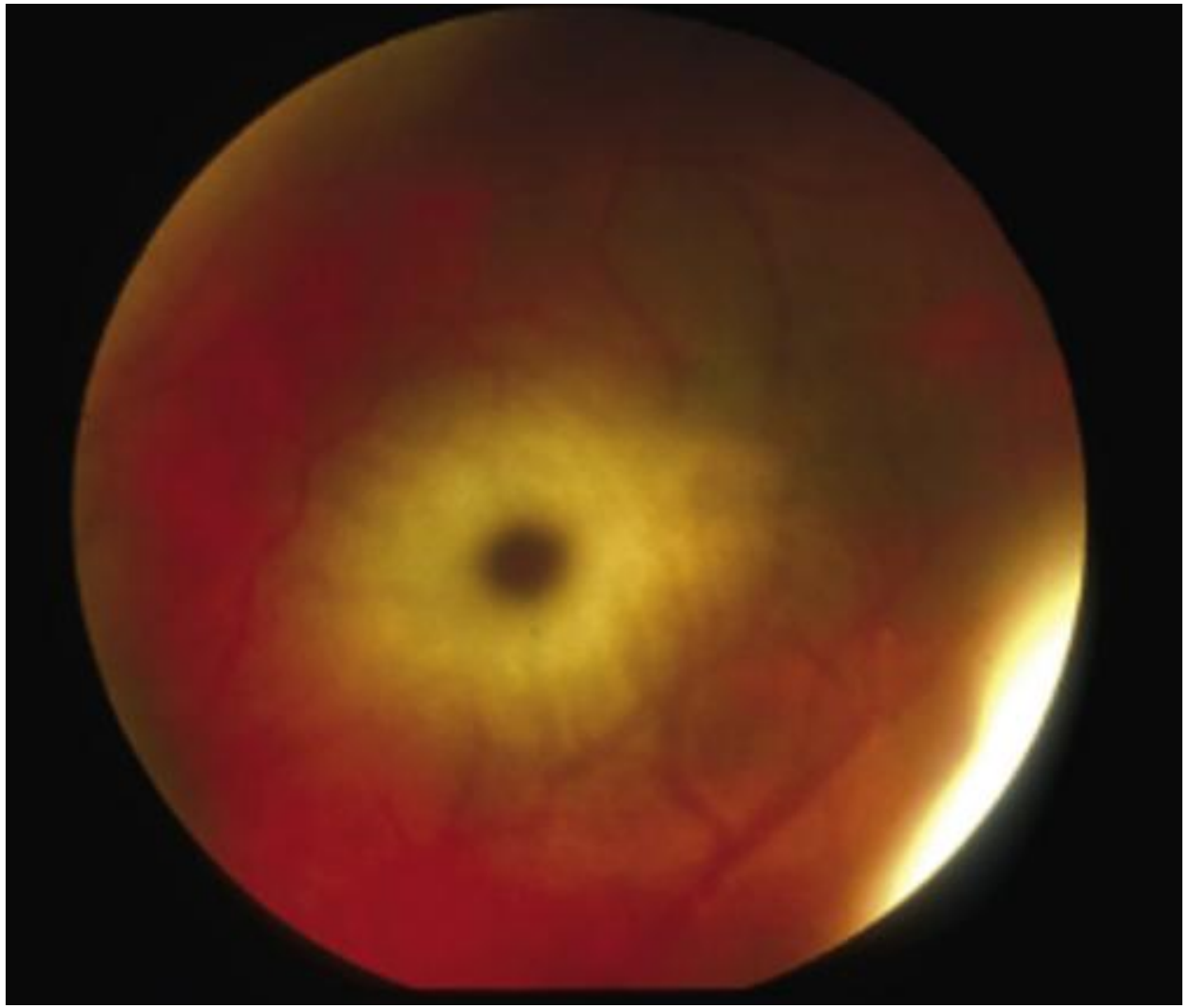
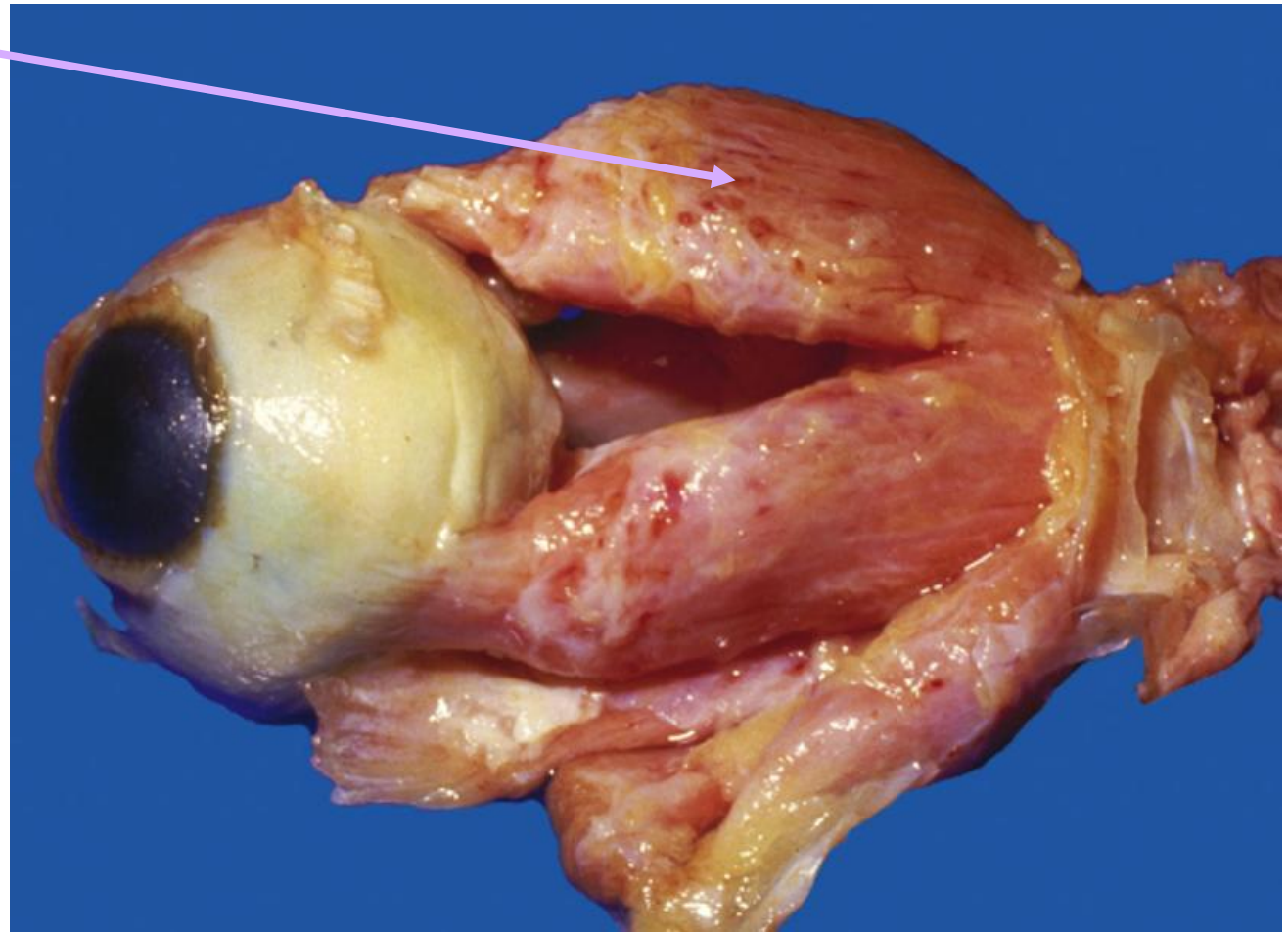


Figure 29.22

Questions

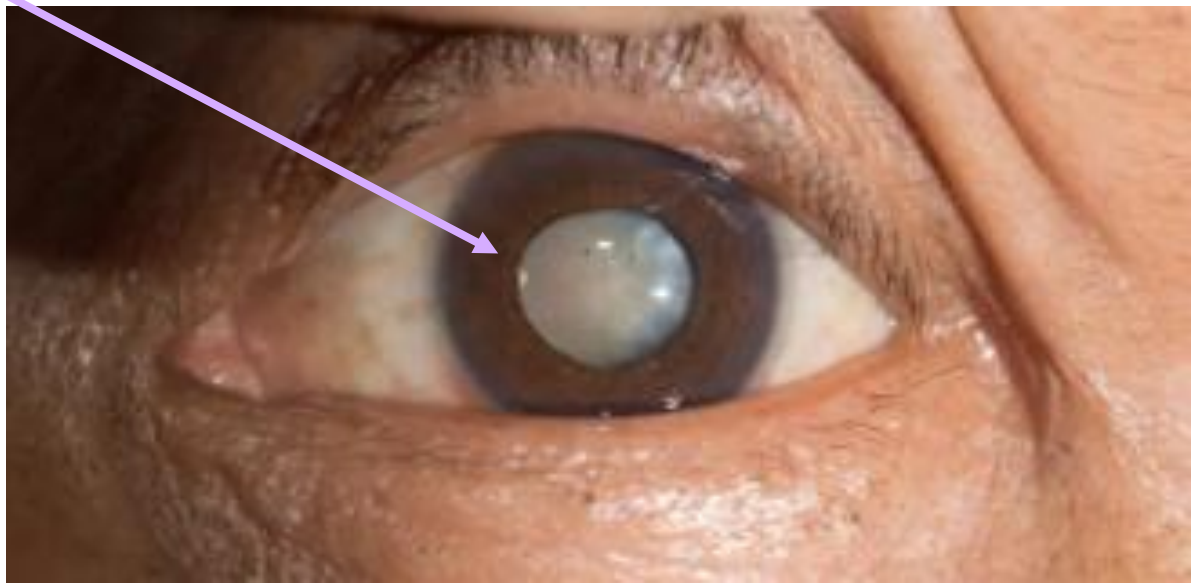
Orbit

- What structures are these?
- The clinical condition of the protruding eye is called?
- What underlying condition causes this?
- What two substances are associated with the extraocular muscle enlargement?



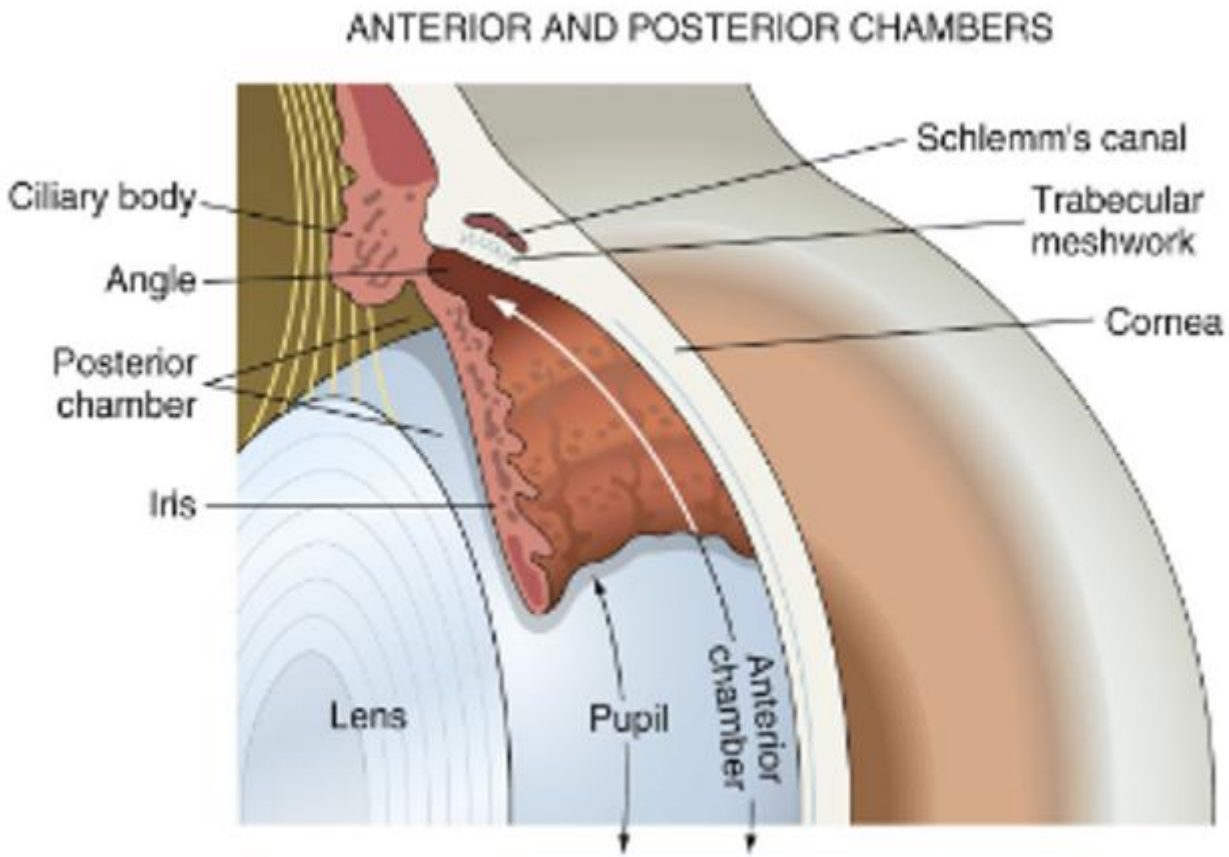
Anterior segment

- What is this?
- What structure of the eye is affected?
- What age do most of these appear?
- What systemic diseases are associated?
- What is the typical treatment?



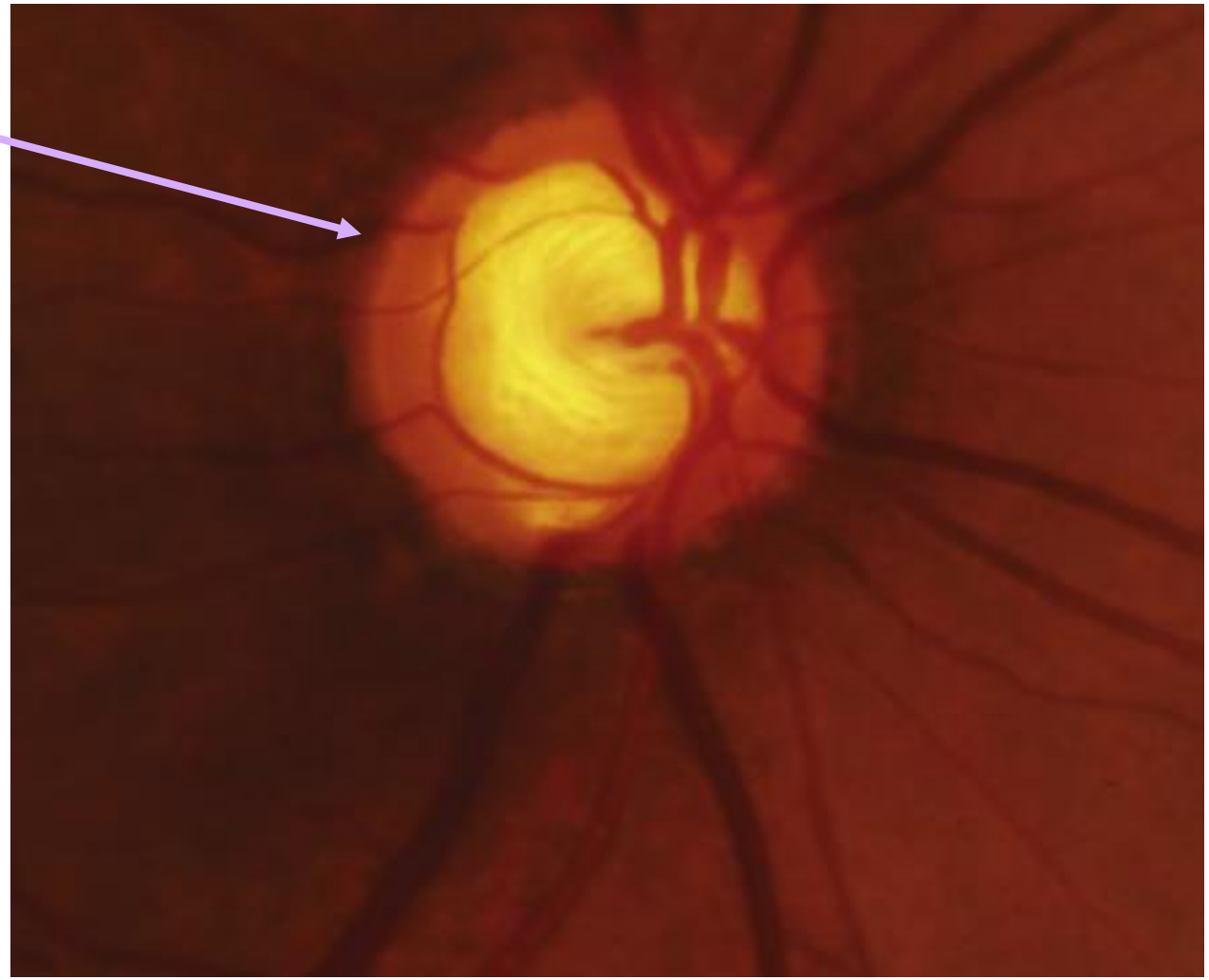
Anterior segment

- Where structure produces aqueous humor and which chamber is it in?
- What two structures are involved in removal of aqueous humor and which chamber are they in?
- What is the name of the group of diseases associated with increased intraocular pressure?
- What are the mechanisms of the 2 major forms?



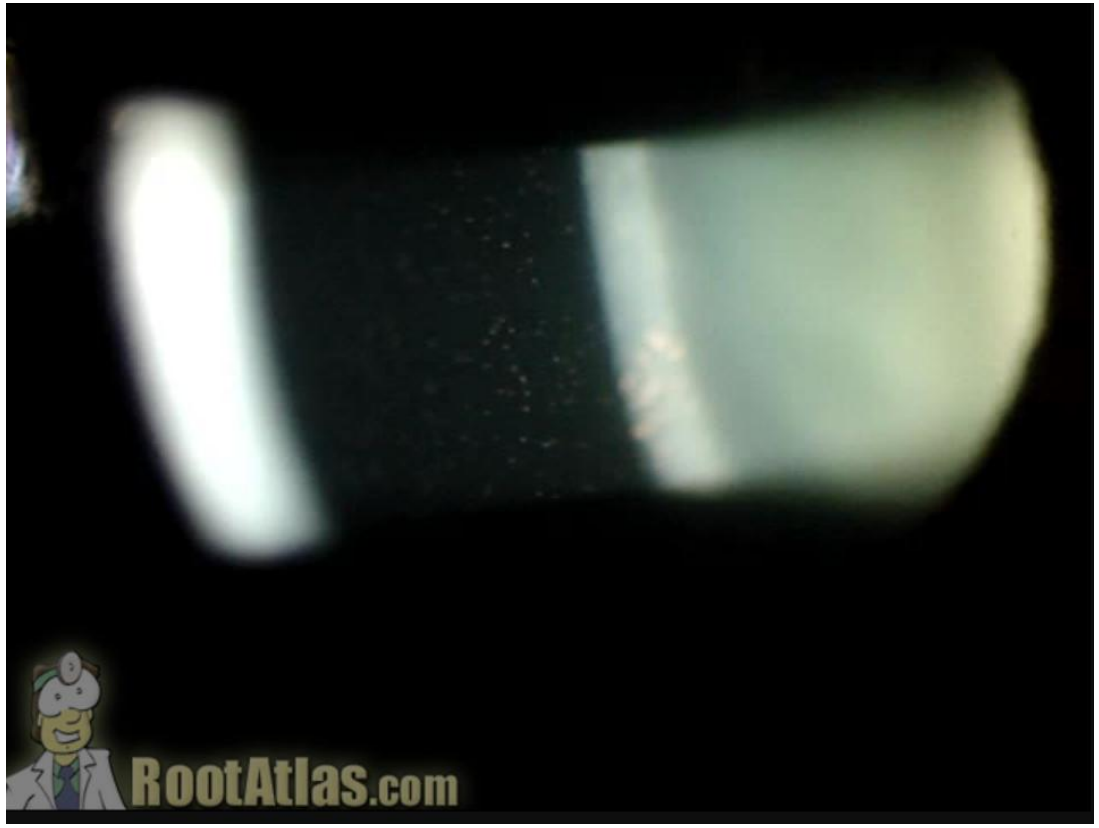
Optic nerve

- What structure is this?
- This is not normal appearance fundoscopically, what is this condition called?
- What causes this condition?
- What underlying eye condition caused this?



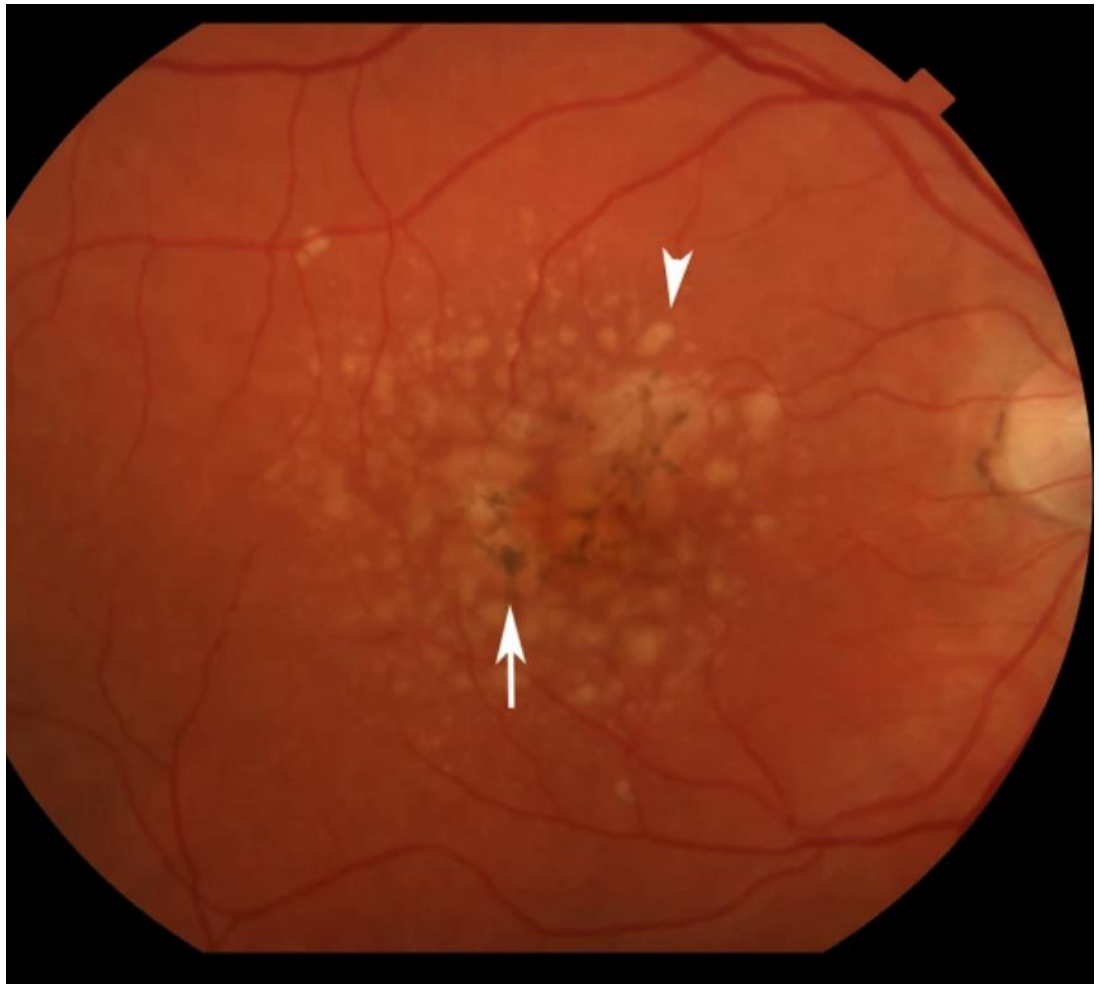
Anterior chamber

- This examination technique is called what?
- What are the particulates called?
- What is the glowing called?
- Name two common non-surgical ocular conditions associated with this finding?
- What underlying conditions are associated with these?



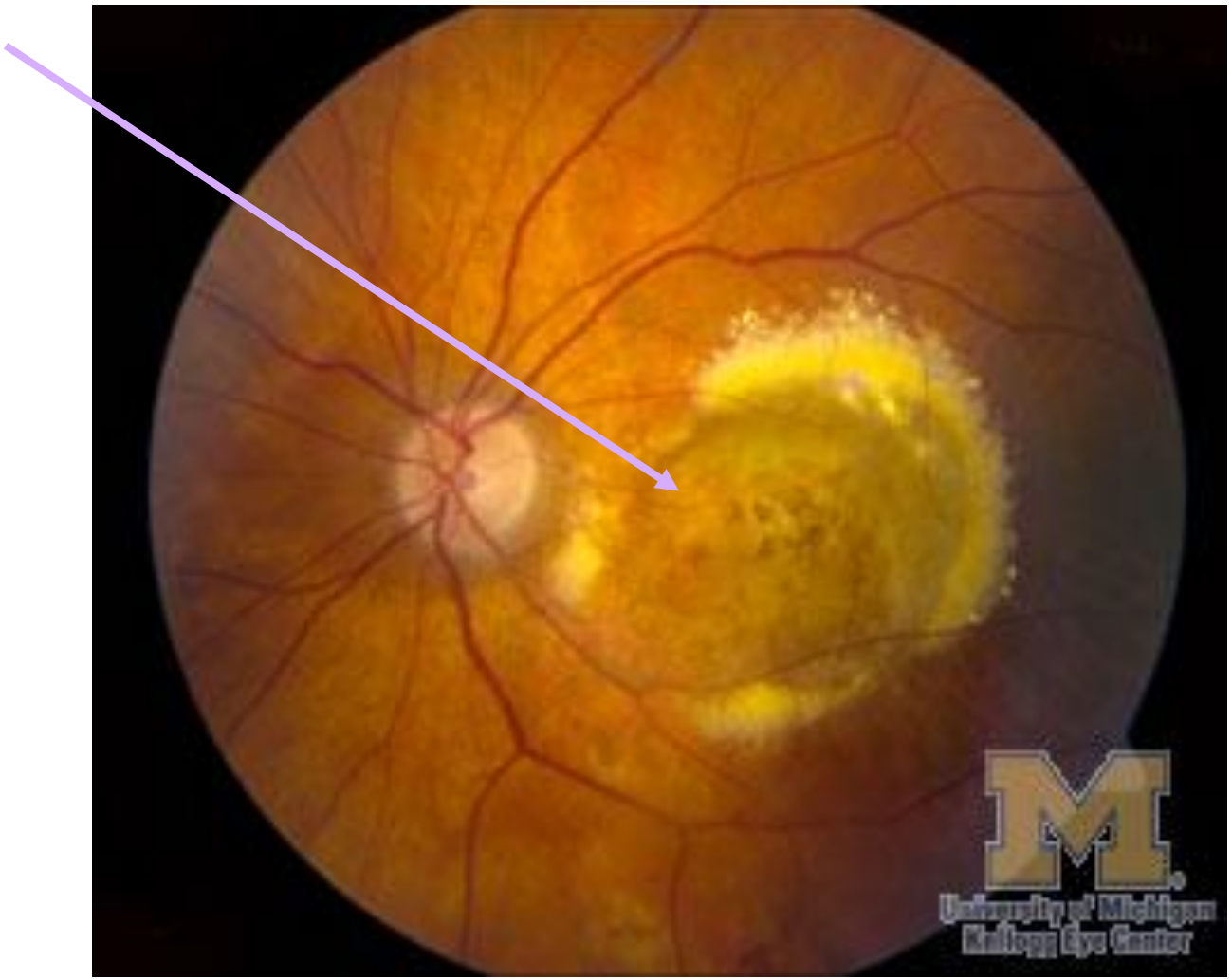
Retina

- What is the arrowhead pointing at?
- What is the arrow pointing at?
- What condition is this?
- Is there a cure for this?



Retina

- What area of the retina is this?
- What is all that yellowish stuff?
- What condition is this associated?
- Is there a treatment?



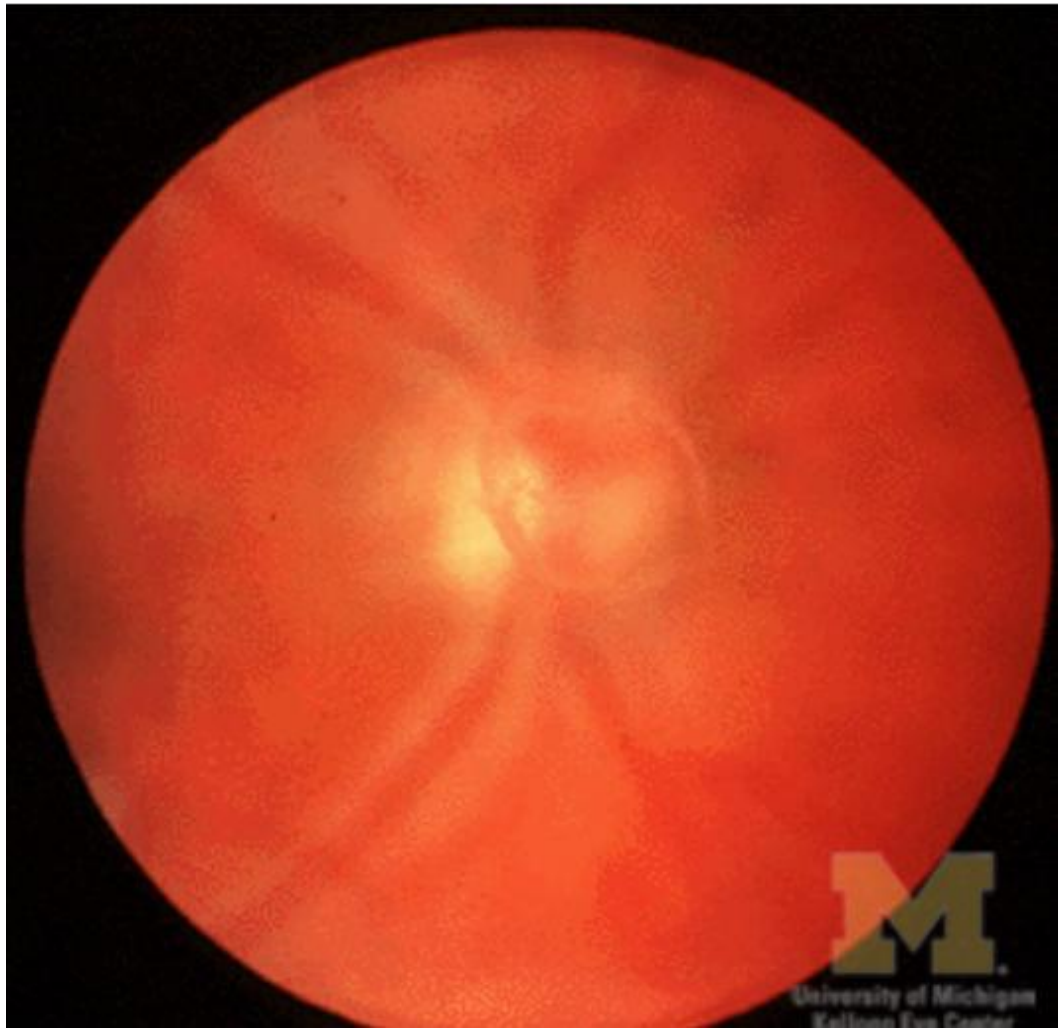
Uvea

- What is this?
- Why is it pigmented?
- What gene mutation is not present in this lesion, that is often seen in its non-ocular forms?
- What is the 5 year survival rate?



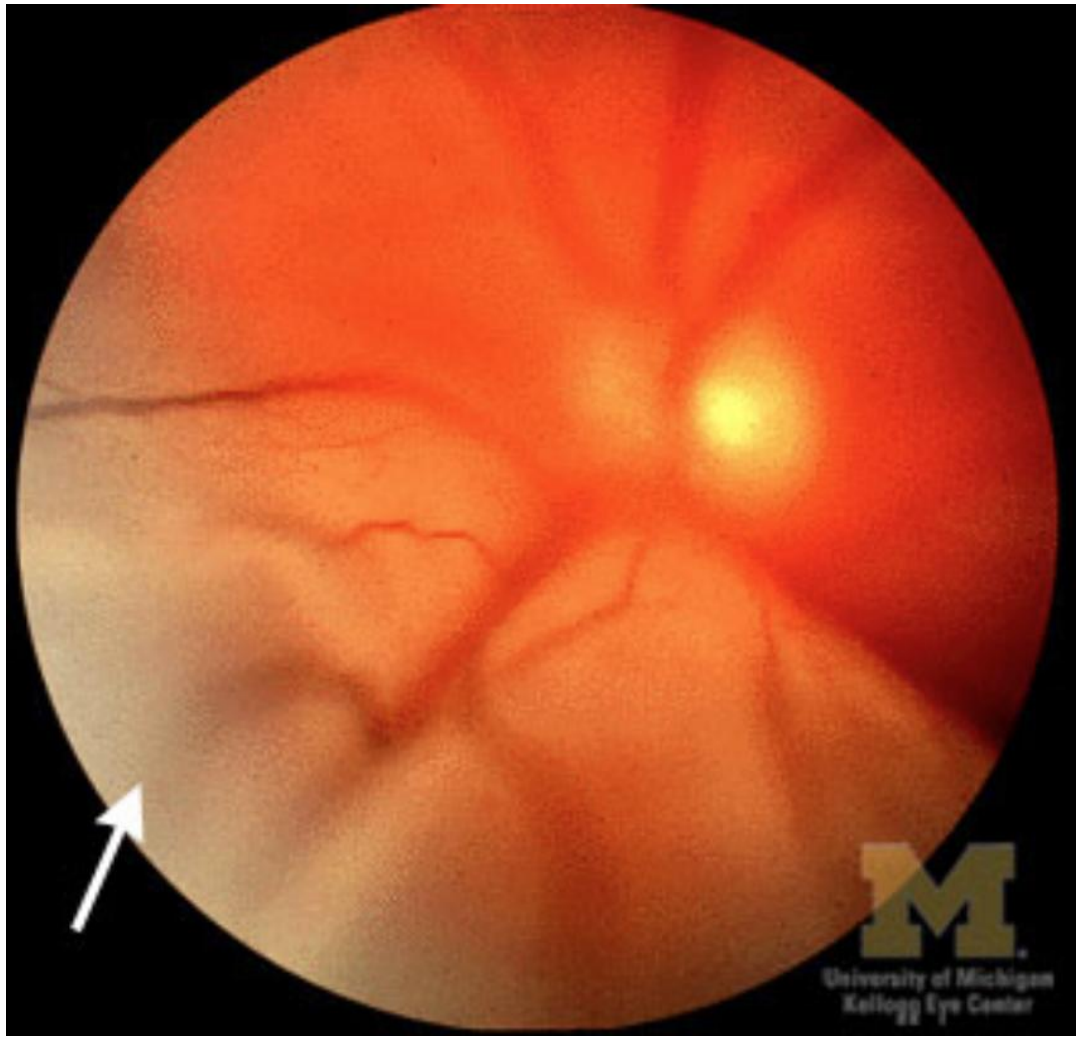
Posterior chamber

- What pathology can be causing this blurry image?
- What are major risks/pathologies if this condition is not treated?



Posterior chamber and retina

- What is the arrow highlighting?
- What is this disease process?
- What is the most common form?



Retina

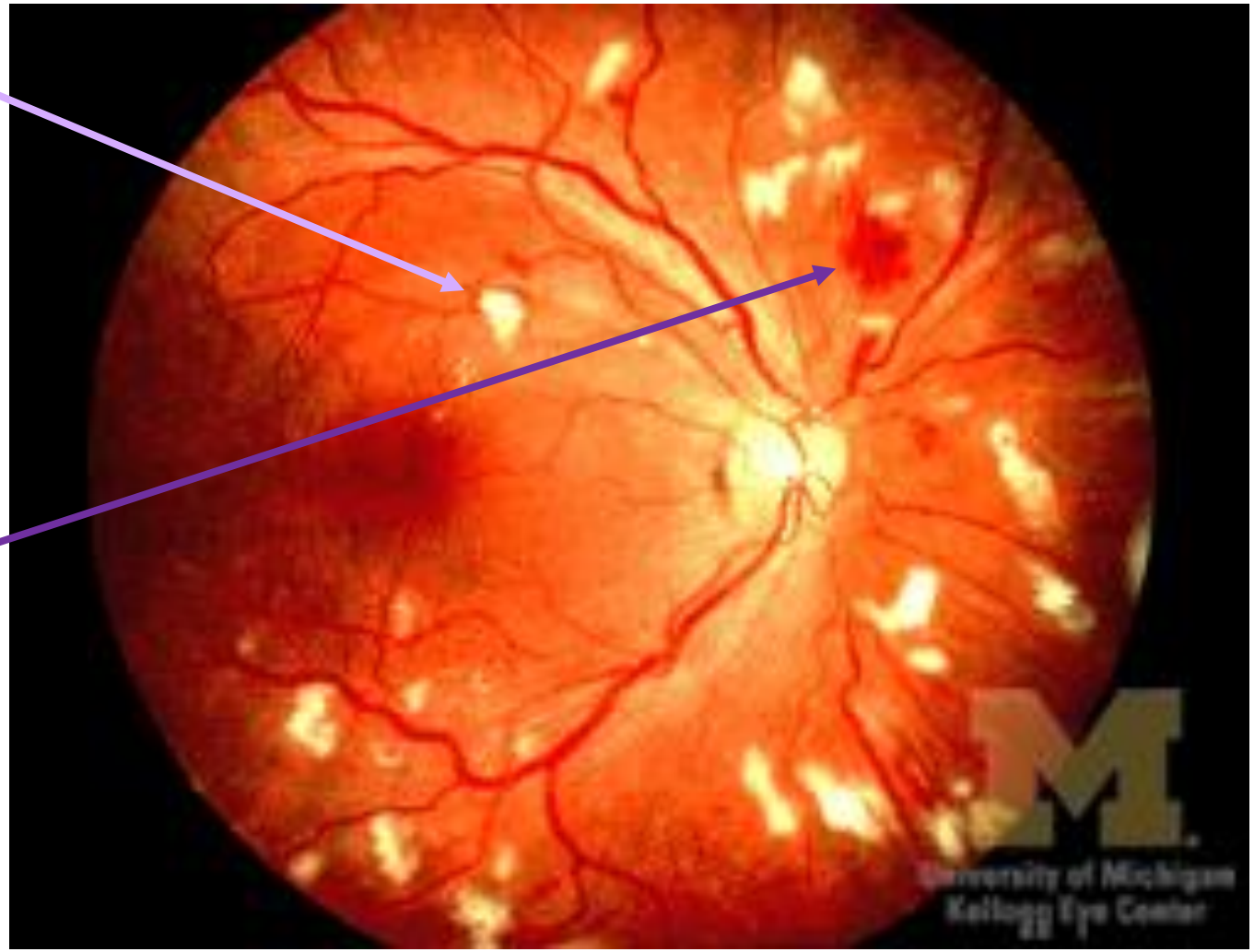
- What is this lesion called?
- What causes it?
- What underlying condition is usually present?
- Why is this a bad thing?



Retina

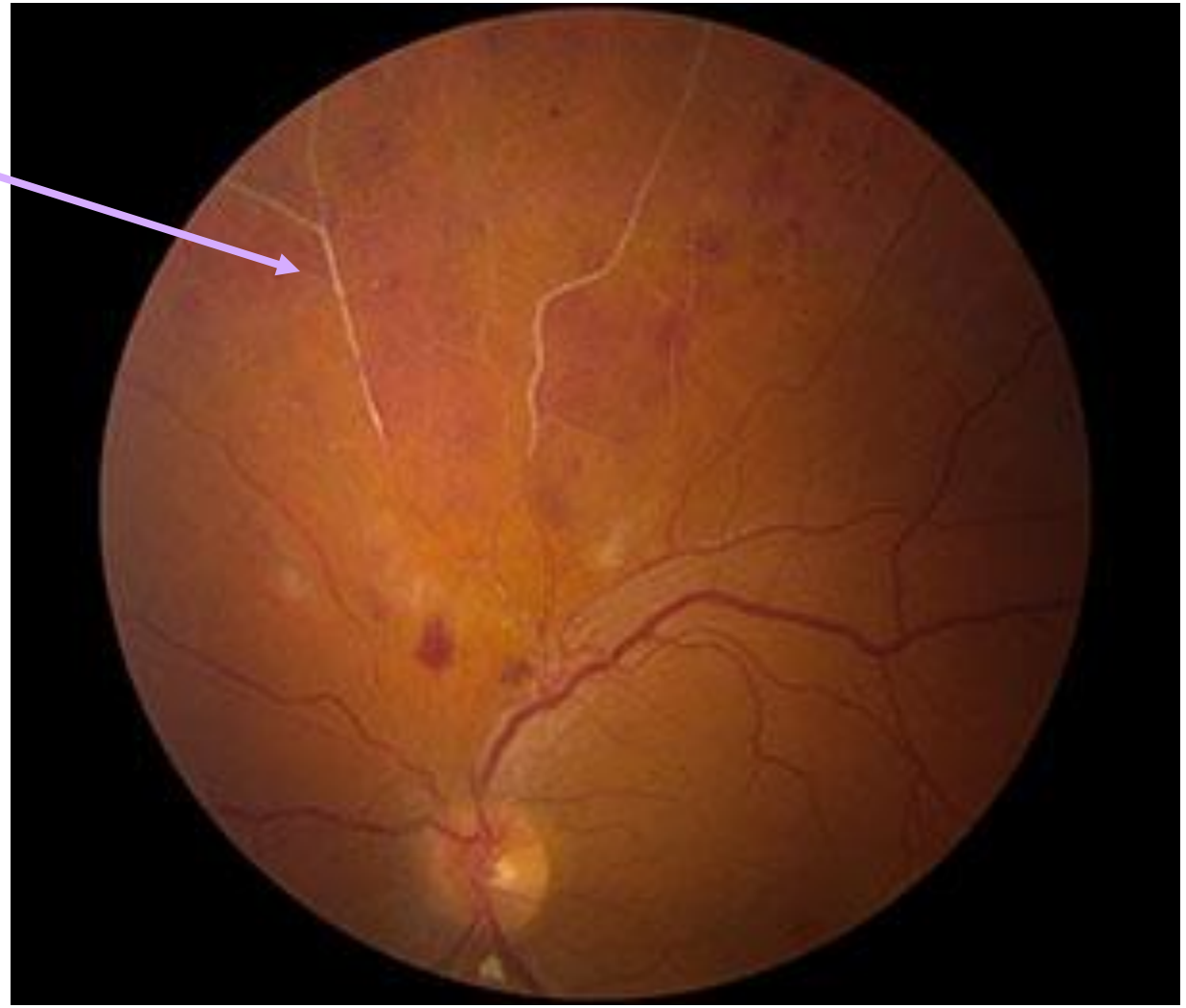
- What are these lesions called?
- What is the underlying ocular pathology causing these?
- What are two important systemic underlying conditions associated with these?

- What are these?



Retina

- What is this called?
- What type of structure is this?
- What histopathology causes this?
- What is the underlying disease process?



Retina

- What structure is this?
- What structure is this and what is the name for it's pathologic change?
- What's happening here?
- What condition are these findings associated with?
- In an emergency situation, what medications should be given?

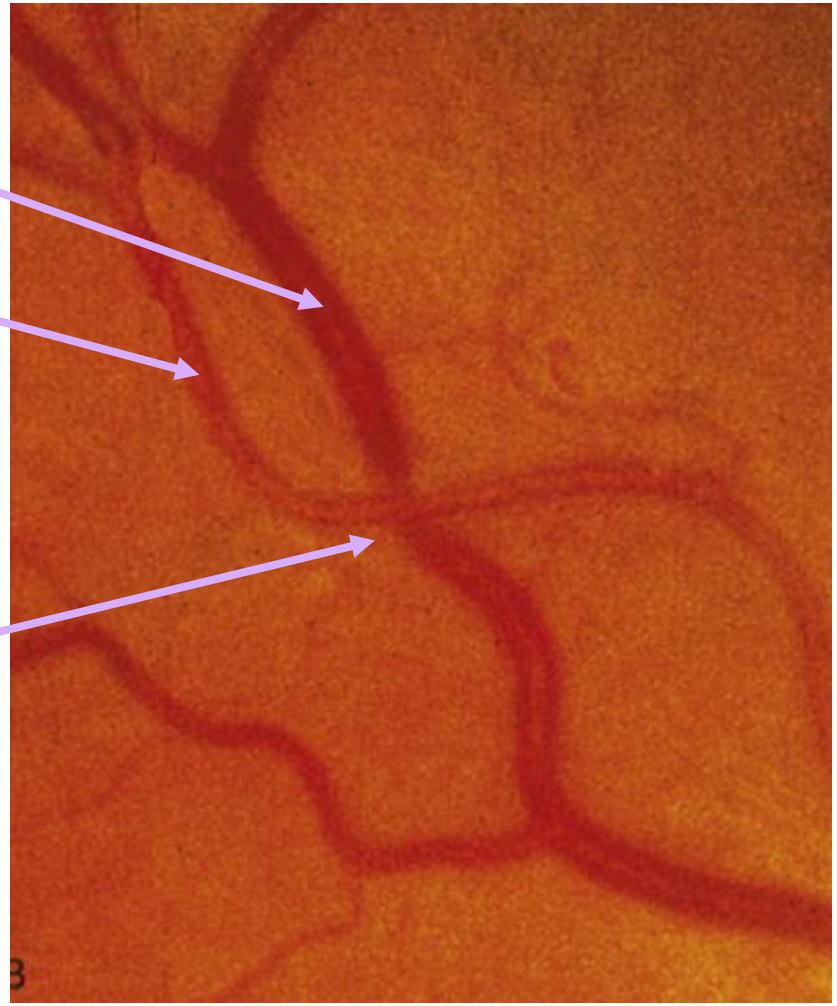
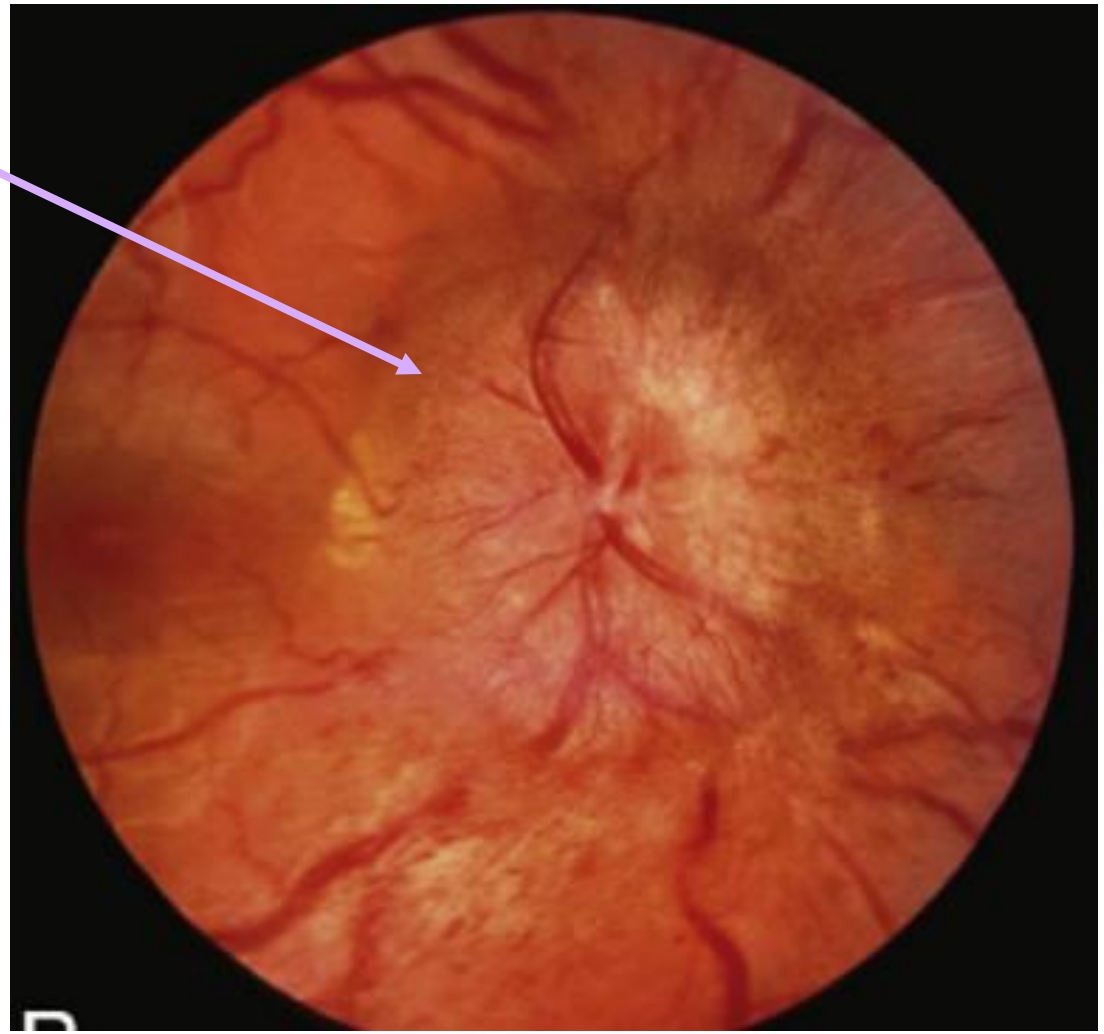


Figure 29.18

Optic nerve

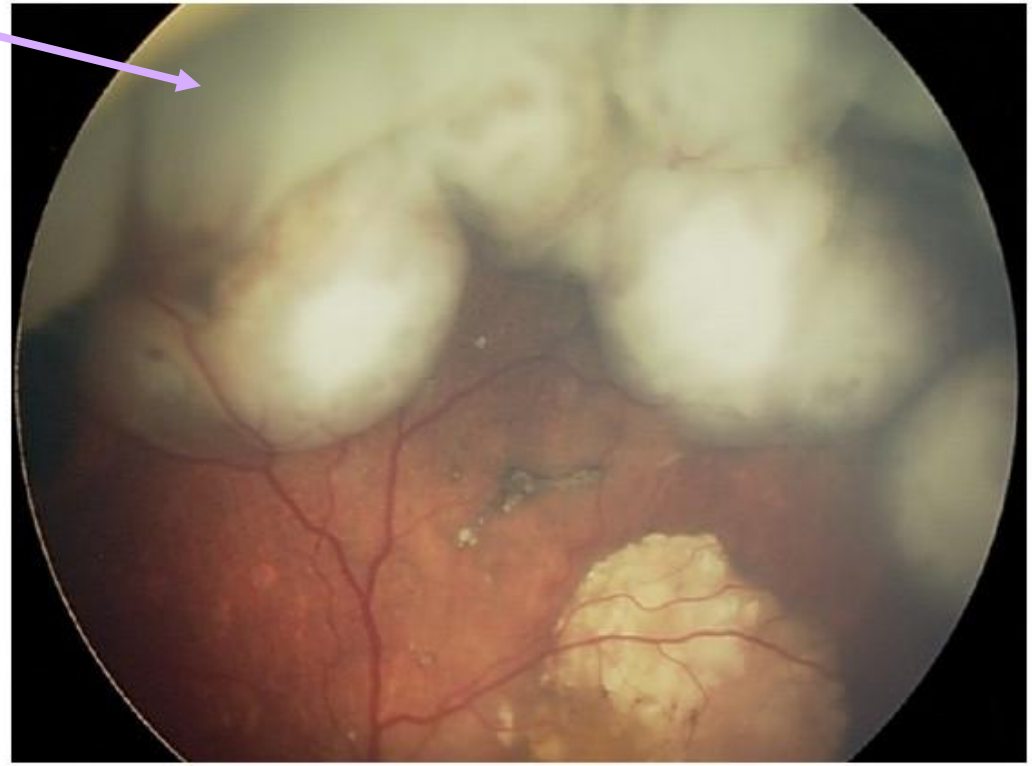
- What structure is this?
- Describe the pathologic lesion to this structure?
- This condition is called?
- What is the underlying cause?
- What is an important initial treatment?



Retina

- What is this?

- In children, this finding is most associated with what disease?
- What gene mutation is this associated with?



Retina

- What is this lesion called?
- What is it composed of?
- What is the underlying disease?

